

ESE 5700

Final Project

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1. Introduction

Static Random-Access Memory (SRAM) is a volatile memory technology widely used in high-performance digital systems due to its low latency, high speed, and compatibility with advanced CMOS processes. The fundamental storage element is the 6-transistor (6T) SRAM cell, which leverages positive feedback through cross-coupled inverters and access transistors to store a single bit of data.

The 6T SRAM architecture presents several design tradeoffs. Stability (read and write margins), performance (access delay), cell area, and power consumption must be balanced carefully. Read stability typically improves with larger pull-down transistors, while writeability improves with smaller pull-up devices—leading to inherent sizing conflicts. Meanwhile, peripheral circuits such as decoders, precharge logic, drivers, and sense circuitry also strongly influence the overall memory performance.

This project implements a full 16×4 SRAM array using a standard 6T memory cell and custom peripheral circuitry. The following sections describe the architecture and verification of baseline and optimized designs, sizing decisions, and performance metrics. The baseline and optimized designs will be compared based on their Figure of Merit (FOM).

$$FOM = Area \cdot Power \cdot Delay$$

Eqn. 1

2. Architecture and Verification of Baseline Design

The baseline architecture of the 64 bit 6T SRAM array consists of the SRAM cells themselves, along with 6 peripheral circuitry blocks.

2.1 Clock Generation

The foundation of the system is a non-overlapping clock generator that converts an ideal input clock into two complementary, non-overlapping phases. The generator consists of two NOR gates followed by two inverters, with final buffered outputs to ensure sufficient drive strength for downstream modules.

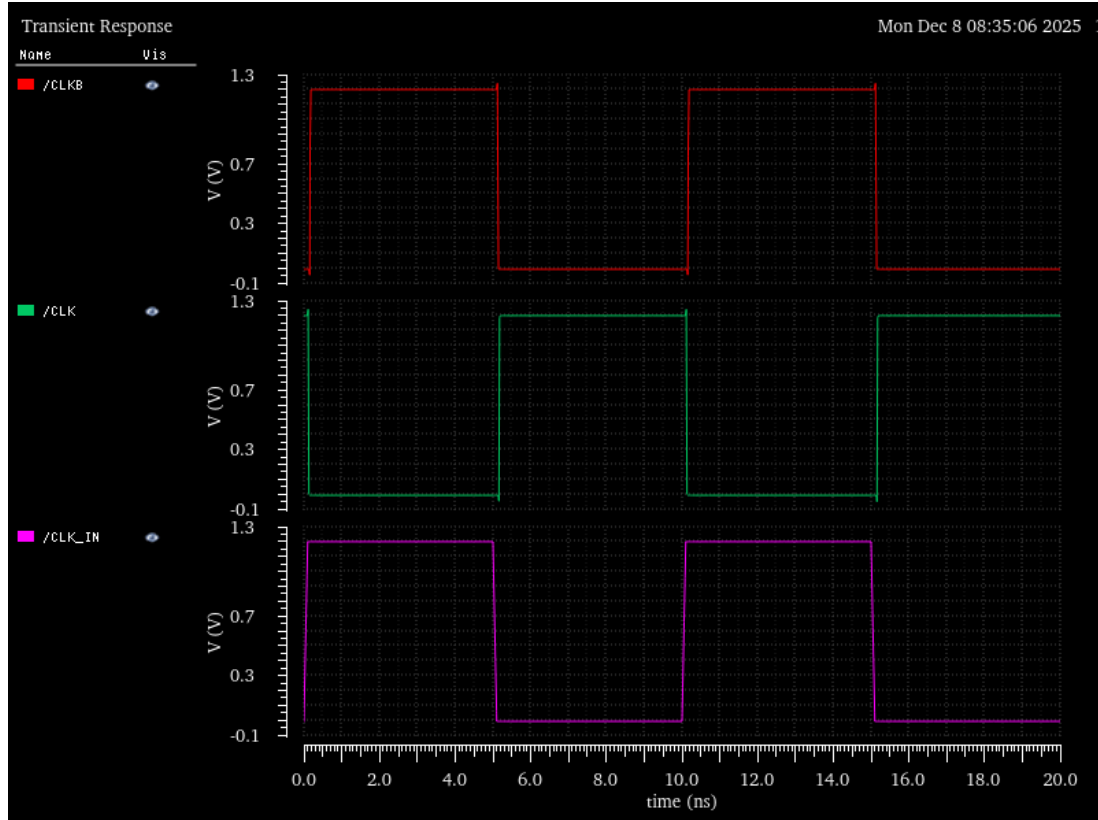


Figure 3. Test output waveforms. CLK and CLKB are clearly opposite in phase.

2.2 Address Decoder

A 4-to-16 row decoder is implemented to generate word-line signals for the 16-row array. The decoder uses a precharged dynamic architecture, allowing fast evaluation during the active clock phase and low power consumption when inactive.

Each clock cycle precharges the internal dynamic node(s); during evaluation the selected address combination discharges one of the 16 word-line outputs, activating the corresponding row.

The decoder architecture and verification are included in Figures 4-7.

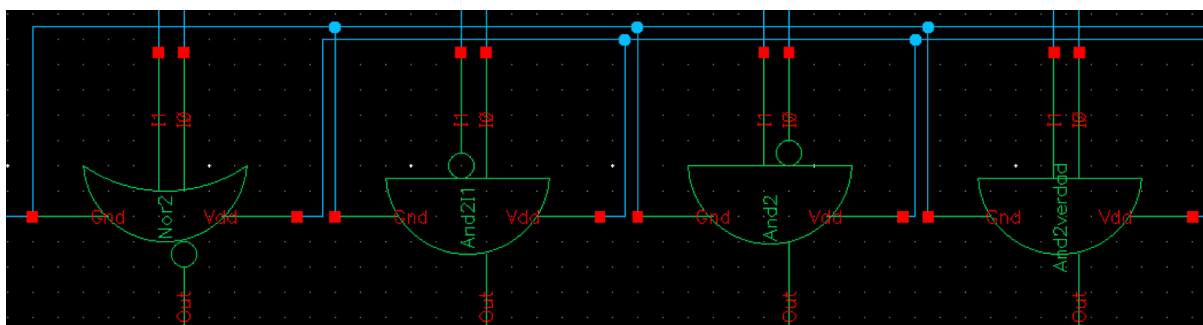


Figure 4. Dynamic decoder combinational logic input.

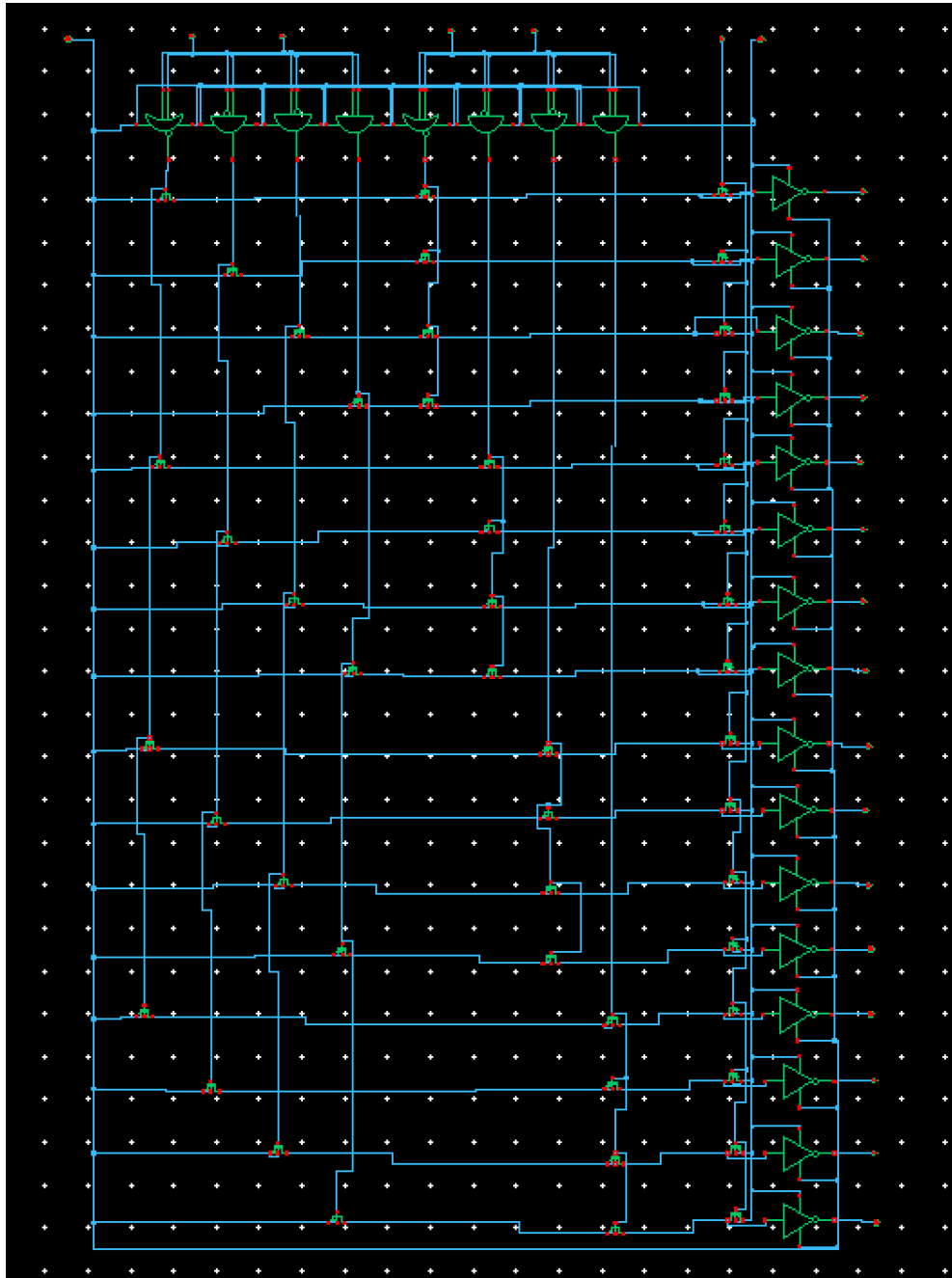


Figure 5. Full decoder architecture. The four bit input entered at the top of the decoder is transformed into the corresponding word line on the right side.

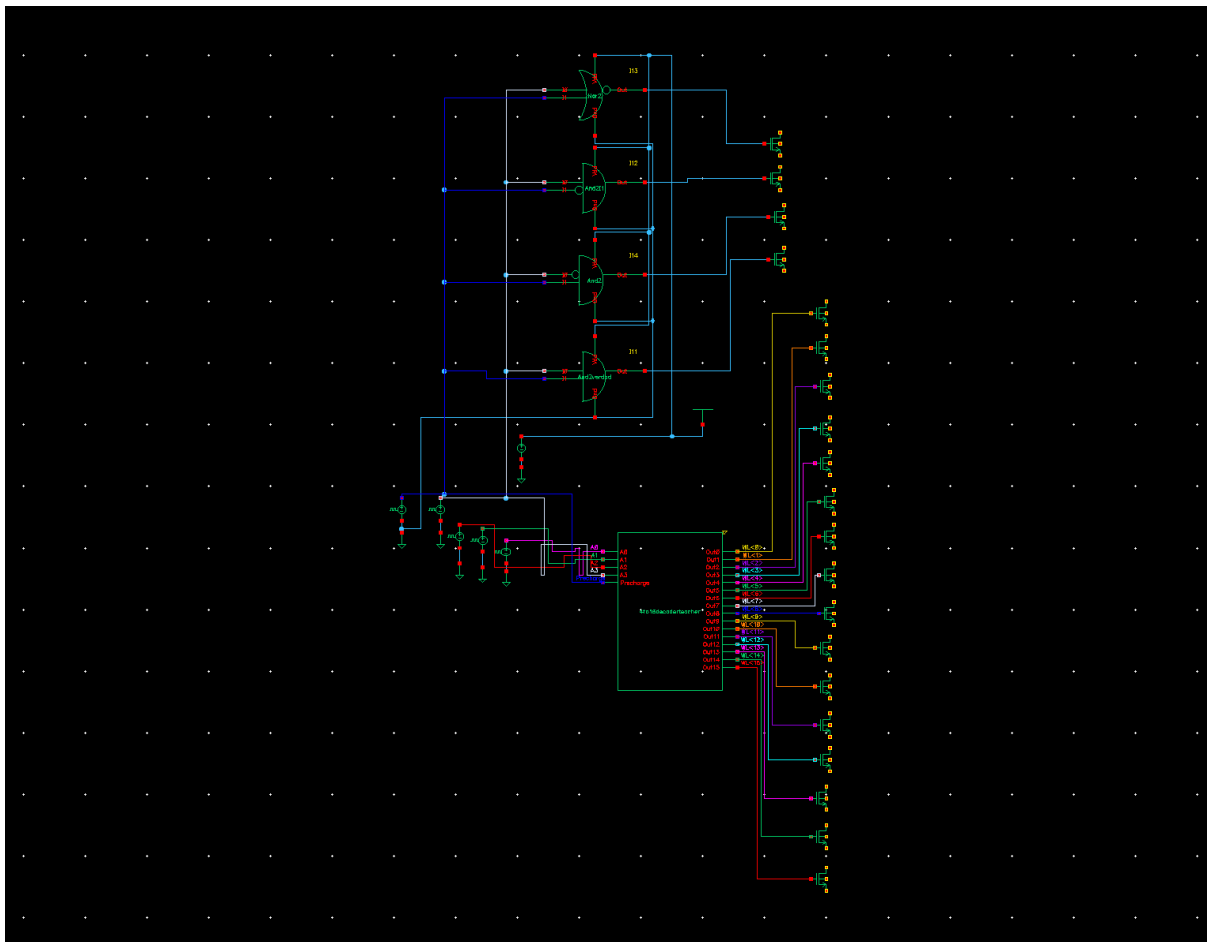


Figure 6. Decoder test schematic. All word line outputs are loaded with transistors of equivalent length to the access pass logic transistors in each SRAM cell.

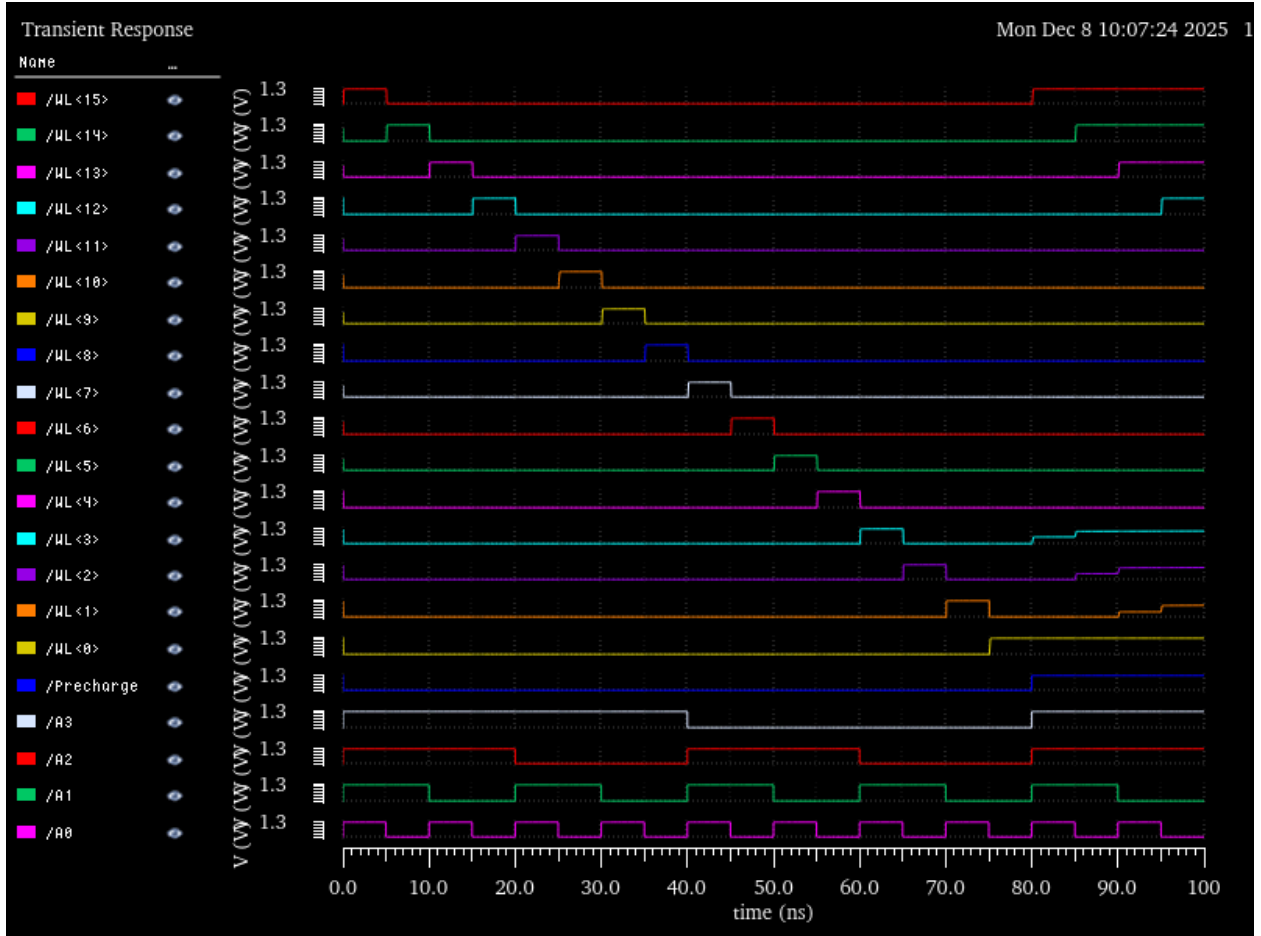


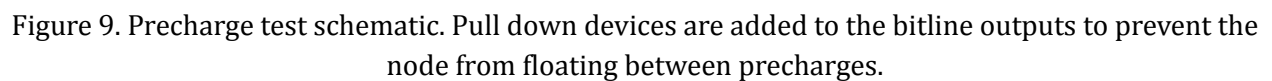
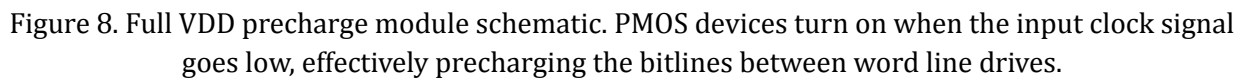
Figure 7. Test output waveforms. A<0:3> corresponds to the address bits, which are cycled through. WL<0:15> are the word lines that the decoder outputs to.

2.3 Bitline Precharge Circuit

Both bitline (BL) and bitline-bar (BLB) are precharged to VDD using a pair of PMOS pass transistors controlled by a clock signal. Precharge occurs only when the clock is low, ensuring that:

- Read and write cycles begin with known, symmetric bitline states.
- Precharge is cleanly isolated from the read/write phases.

This simple VDD precharge method provides consistent read speed. The precharge module schematic and verification are included in Figures 8-10.



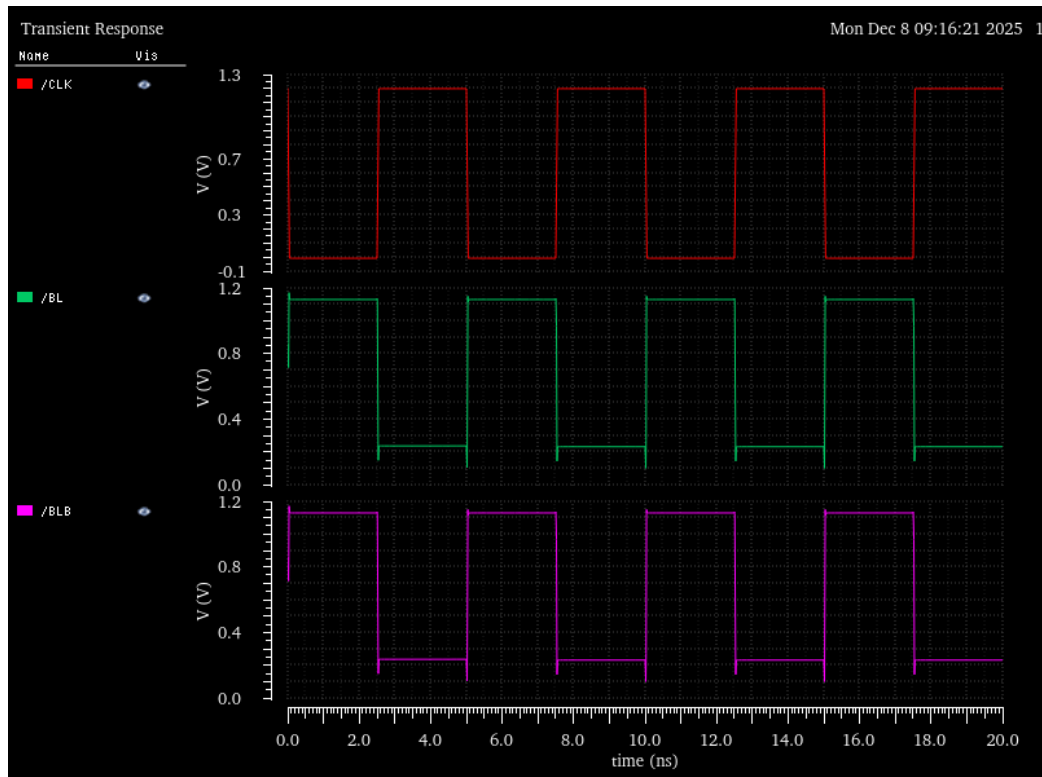


Figure 10. Test output waveforms. Whenever the clock signal is low, the precharge activates, bringing both `BL` and `BLB` to `VDD`.

2.4 Write Enable Module

The write driver receives `Data` and `Write Enable` inputs and drives the differential bitlines accordingly. It includes:

- A pass NMOS driven by `Data`
- An inverter generating `Data-bar`, driving a second pass NMOS
- `Write Enable` controlling series NMOS devices on both `BL` and `BLB`

When `Write Enable` is asserted, the selected row receives a strong differential drive to write the intended value into the SRAM cell. When `Write Enable` is low, the bitlines remain undriven by this module.

Figure 11 clearly shows the write module architecture. The transistors connected to the bitlines are sized up to overcome the large bitline capacitance.

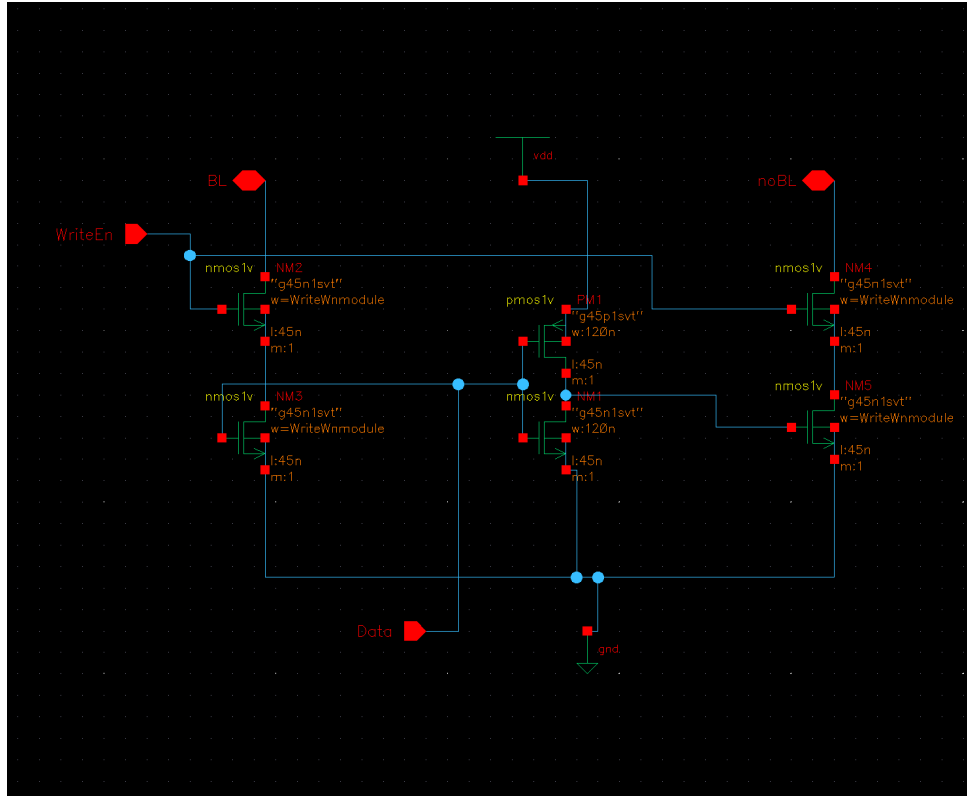


Figure 11. Write module schematic. Data gets pushed to the bitline and the inverse of data is pushed to bitline bar. Write enable connects/disconnects the circuit from the bitlines.

Verification of the write module's operation will be shown through the SRAM cell verification.

2.5 Tri-State Buffer

The tri-state buffer provides conditional signal propagation based on an Enable input. The input first passes through an inverter; its output drives a PMOS/NMOS pull-up/pull-down pair. The Enable signal controls additional gating transistors in series with these devices, and its inverted form drives the complementary gating device.

When Enable = 1, the module behaves as a standard buffer; when Enable = 0, the output floats, providing high-impedance isolation.

Simulated waveforms confirm correct propagation when enabled and stable high-impedance behavior when disabled.

Figures 12-14 show the tri state buffer schematics as well as these waveforms.

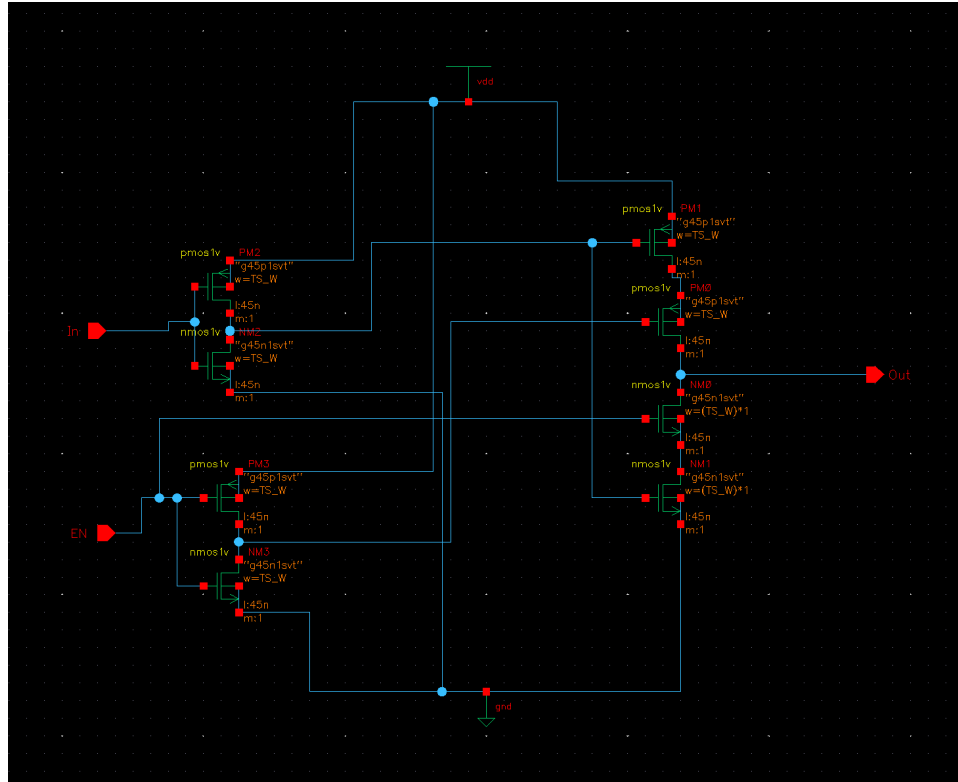


Figure 12. Tri state buffer module schematic. Output follows input while Enable is high. Otherwise, the output is a high impedance node.

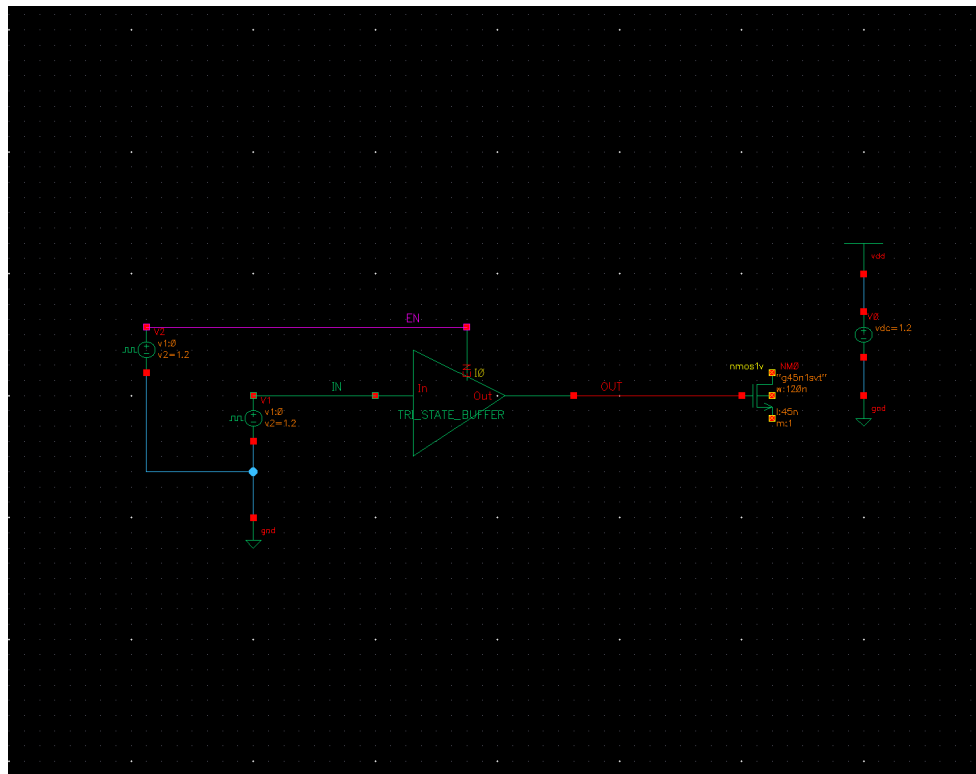


Figure 13. Tri state buffer test schematic.

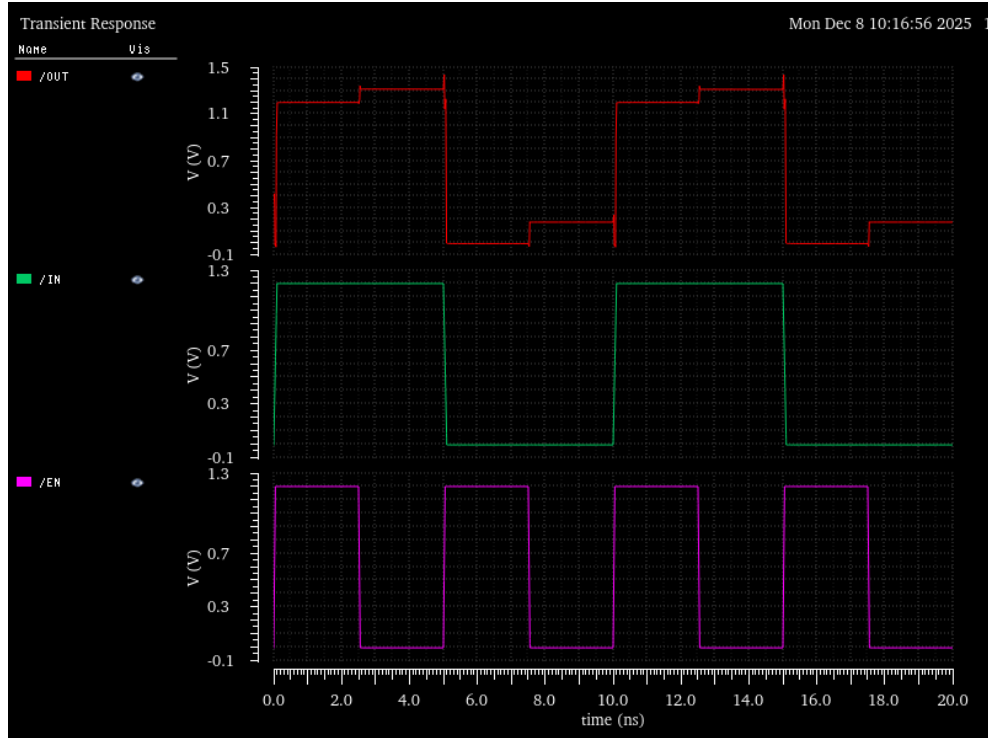


Figure 14. Test output waveform for tri state buffer. When EN is high, the output directly follows the input. When EN is low, the output is floating, which is represented by a drift to a higher voltage.

2.6 Clocked Transmission Gate with Output Pulldown

To ensure deterministic timing and eliminate floating internal nodes, a clocked transmission-gate pull-down (TG-PD) circuit is used. The structure consists of complementary pass transistors gated by CLK and CLKB, along with a pull-down NMOS whose gate is tied to CLKB. When CLK is high, the transmission gate becomes transparent and passes the input signal to the output. When CLKB is high (i.e., during the low phase of CLK), the pull-down device forces the output to ground, preventing charge sharing and guaranteeing well-defined logic levels. This ensures that all control signals within the array follow clean clock boundaries without unintended residual charge. Figure 15 shows the transmission-gate schematic, and Figure 16 shows the test configuration. The simulation results in Figure 17 demonstrate transparent behavior during CLK high and consistent discharge to ground during CLK low, with no floating nodes or half-select behavior observed during verification.

For the TG-PD control path, the transmission-gate devices are sized to $5\times$ minimum to ensure that critical control signals—such as decoder outputs, write-enable, and read-enable—propagate with minimal delay and minimal voltage degradation. The larger transmission gate provides a low-resistance path for both rising and falling edges, preserving signal integrity as the signals drive downstream word-line and peripheral circuitry. The dedicated pull-down device is sized to $6\times$ minimum to guarantee fast and reliable discharge of the control node when the signal is

de-asserted, preventing glitches or partial activation of memory elements. This sizing strategy ensures that the TG-PD circuit delivers clean, full-swing, clock-aligned control signals while maintaining reasonable power and area.

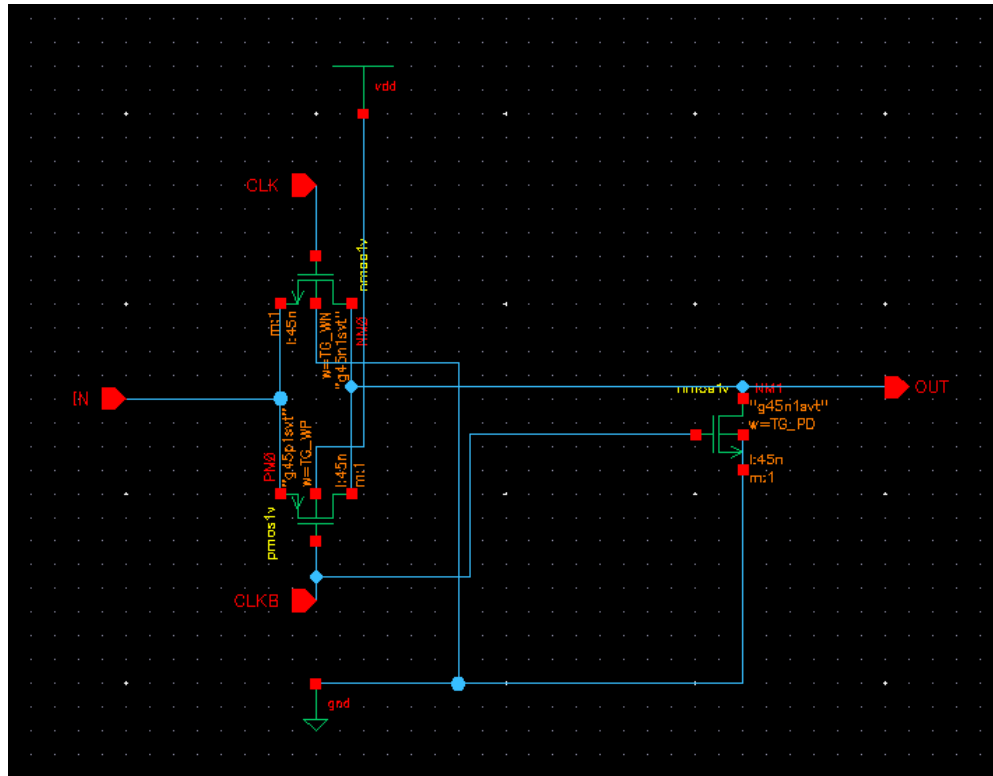


Figure 15. Transmission gate with CLKB activated pull down NMOS. Synchronizes SRAM periphery signals and pulls down node during precharge to avoid corrupting data.

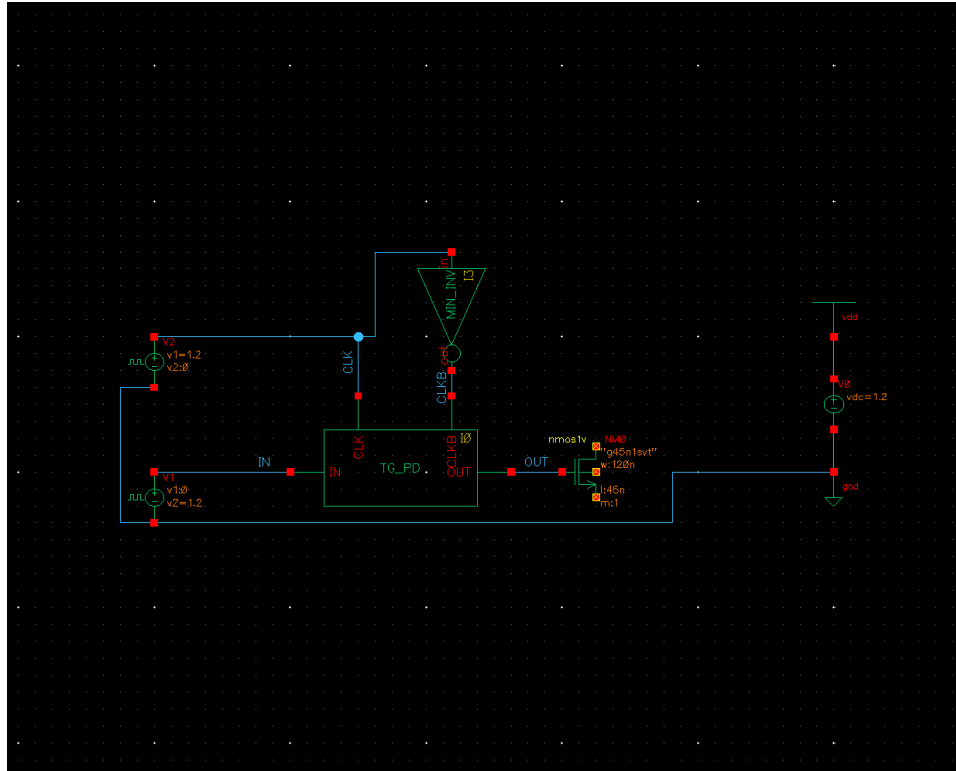


Figure 16. Transmission gate test schematic.

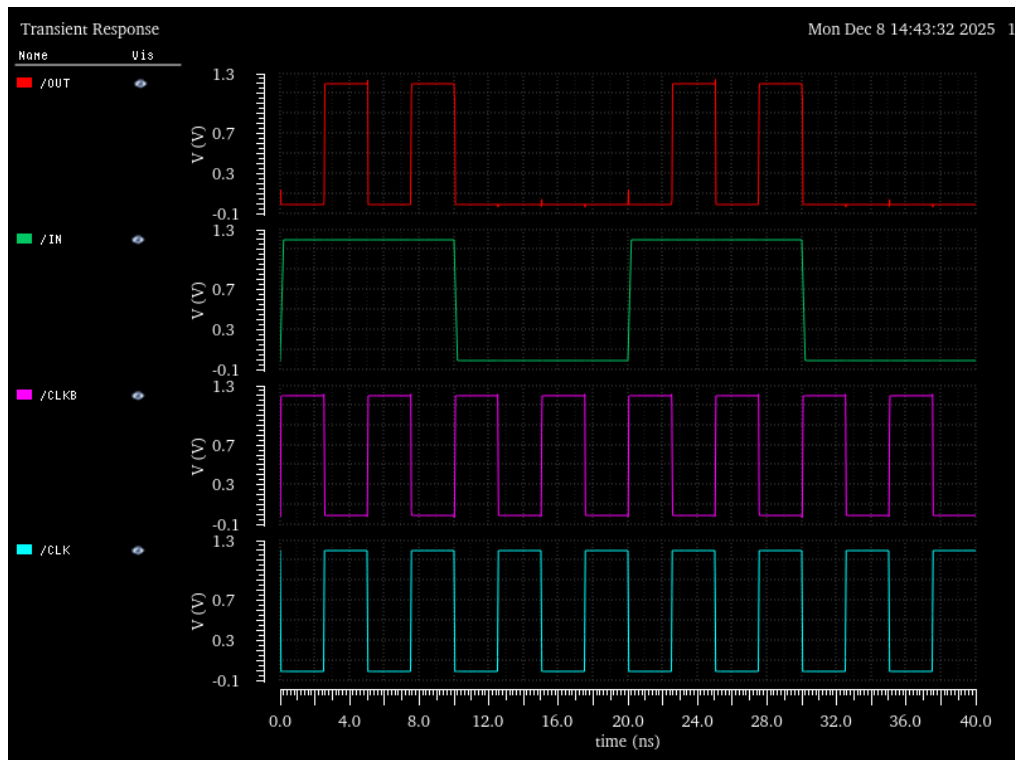


Figure 17. Test output waveforms for TG PD cell. Output follows input on CLK high and gets pulled down to GND on CLKB high.

2.7 6T SRAM Cell

The storage element is a standard 6-transistor SRAM cell composed of two cross-coupled inverters and two access NMOS transistors controlled by the word line. The cell supports both read and write operations depending on the state of the word line and bitlines.

Figure 18 includes the full schematic of the SRAM cell. Each transistor width is left as a variable. In the baseline 6T SRAM cell design, the pull-up, access, and pull-down devices will be intentionally sized to balance writeability, read stability, and cell area. The pull-up PMOS is kept at minimum width to weaken the feedback inverter during a write, allowing the bitlines and access transistors to more easily flip the stored value. The access transistors are sized to 2x minimum to ensure sufficient drive strength for both read and write operations while avoiding excessive bitline loading. The pull-down NMOS is upsized to 8x to strengthen the storage node during a read, preventing the internal node storing “0” from being inadvertently disturbed when the bitline rises toward VDD. This sizing strategy reflects the classic tradeoff between write margin and read static noise margin, where a strong pull-down and weak pull-up favor read stability, and moderately sized access devices preserve acceptable write performance without compromising cell robustness.

In the optimized design, the transistors will be minimum size, and the peripheral circuitry will be modified to minimize Figure of Merit (FOM).

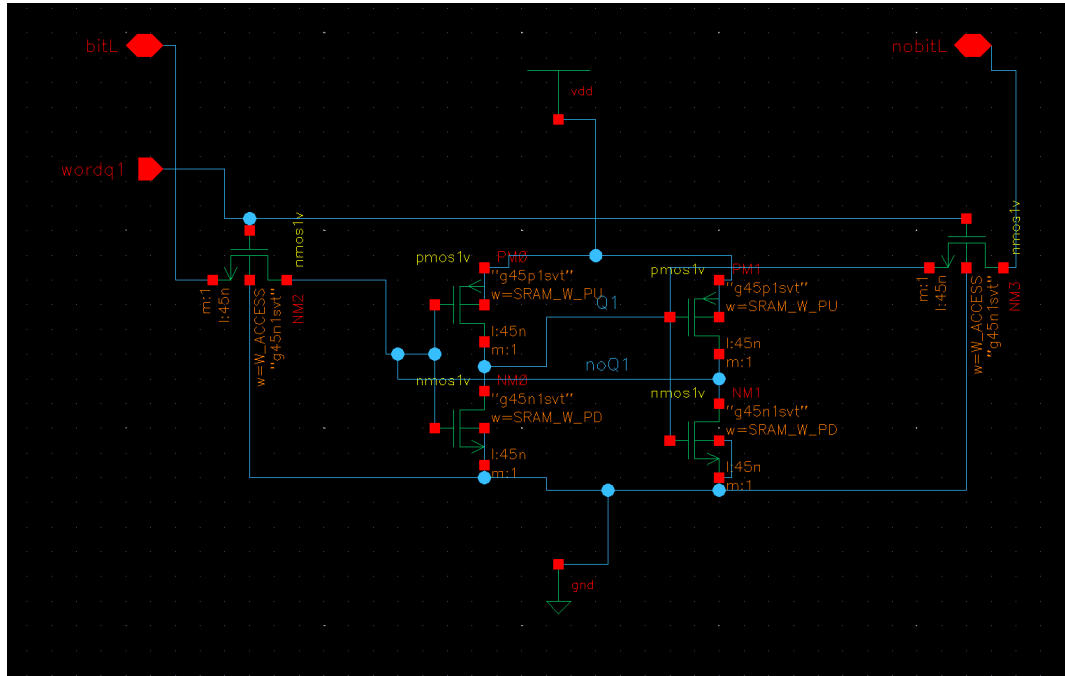


Figure 18. Standard 6T SRAM cell. Data is stored in the cross coupled inverter pair.

The 6T SRAM cell was connected to all peripheral circuitry and tested to verify functionality. Figure 19 includes the test schematic for the cell, and Figure 20 includes the resultant waveforms.

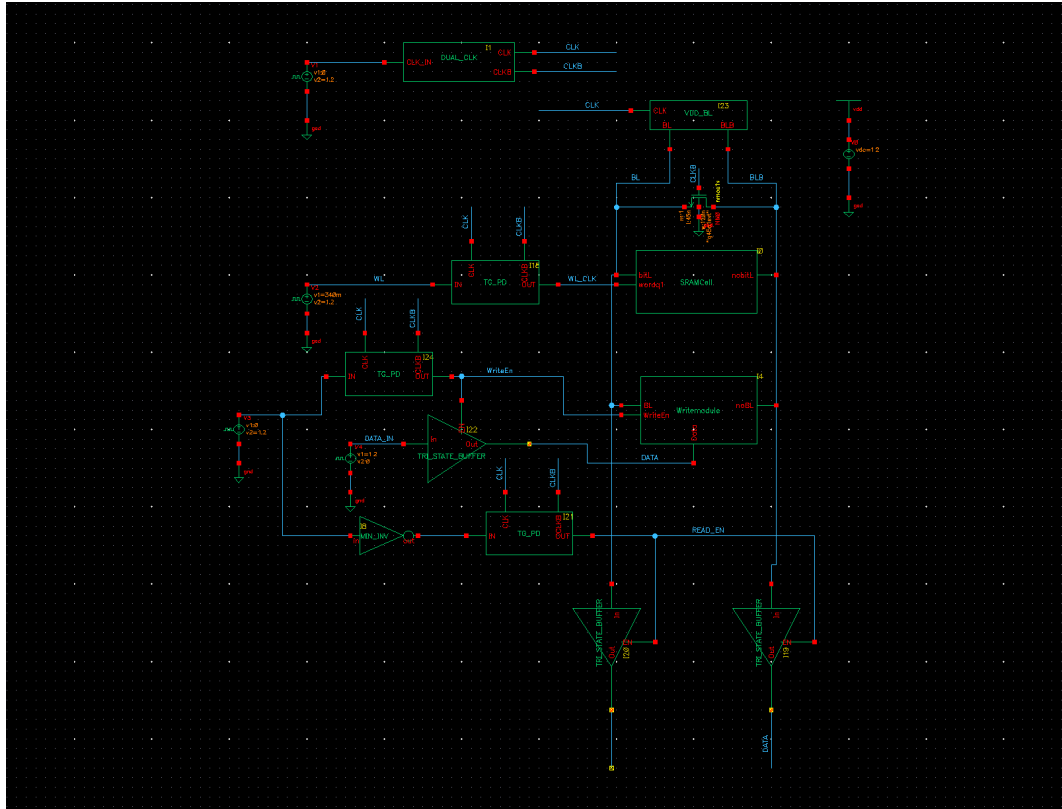


Figure 19. Fully clocked test schematic for 6T SRAM cell

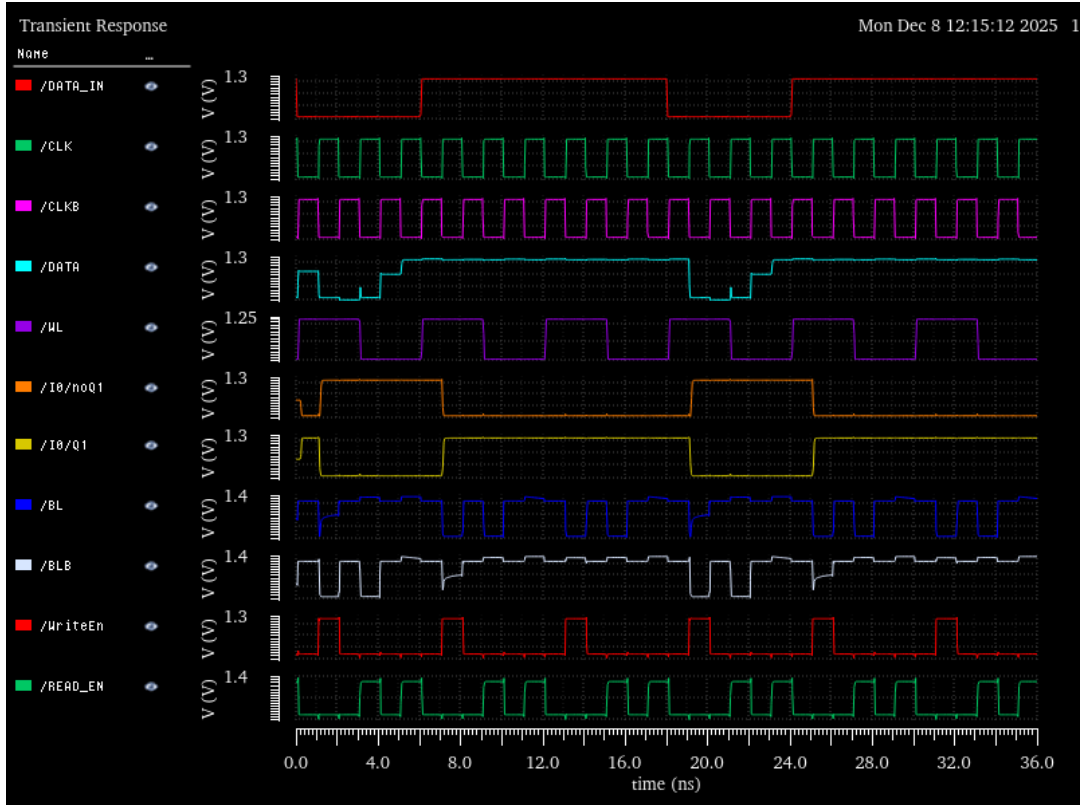


Figure 20. Test output waveforms for a single, baseline SRAM cell.

2.8 SRAM Column Architecture

Each SRAM column consists of 16 vertically stacked 6T SRAM cells sharing a common pair of differential bitlines (BL and BLB). Although the bitlines run continuously through the column, each row has its own dedicated word line, allowing independent read/write access to any of the 16 cells.

During a read operation, the selected cell couples its internal node to BL and BLB, creating a small differential voltage that propagates up the column. During a write operation, the bitlines are driven strongly by the write driver, overpowering the cross-coupled inverters of the selected cell.

This column-level structure forms the fundamental building block of the full array. The column schematic is shown in Figure 21.

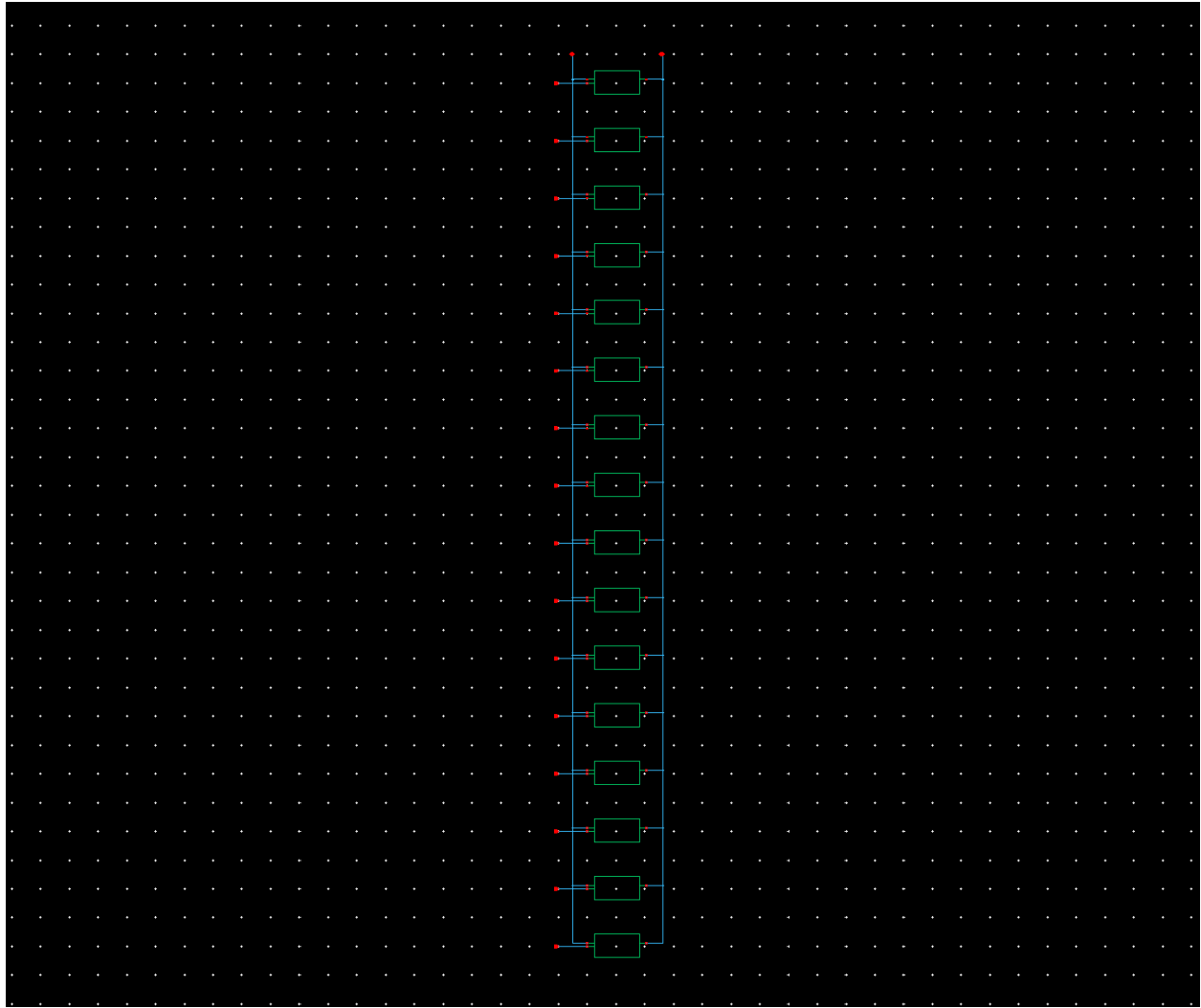


Figure 21. 16 SRAM cells oriented in a column. Each cell shares both bitlines. Each row has a different word line.

Simulations of the column and the full array are essentially the same, and verification of this architecture will be provided in the next section.

2.9 Full 64-Bit SRAM Array Architecture

The full memory array is constructed from four SRAM columns, each implemented as described above. All four columns share a common set of 16 word lines produced by the 4-to-16 dynamic decoder. The resulting structure provides a complete 64-bit word (4 columns $\times$ 16 rows).

To maintain proper timing and prevent half-selected behavior, the decoder outputs are not connected directly to the word lines. Instead, each decoder output feeds a 16-input / 16-output transmission-gate with pull-down (TG-PD) block, which performs two critical functions:

1. Clocking the word lines:

The TG-PD passes the decoder output only during the high phase of CLK, ensuring that memory accesses occur in well-defined cycles driven by the non-overlapping clock generator.

2. Grounding the word lines when inactive:

During CLKB high (CLK low), the pull-down NMOS forces each word line to 0, preventing floating states and eliminating leakage-induced corruption.

All columns share a full VDD bitline precharge implemented by PMOS transistors gated by the clock. When CLK is low, BL and BLB across all columns are charged to VDD.

In addition, each column includes an NMOS equalizer between BL and BLB, controlled by CLKB. During CLKB high (the precharge phase), this equalizer equalizes the two bitlines to eliminate any residual differential from the previous operation, ensuring consistent read behavior.

A TG-PD stage also gates the Write Enable signal, ensuring that WE is only high during the active (CLK high) portion of the cycle. This prevents accidental writes during precharge.

The data input is passed through a tri-state buffer before entering the write driver. The enable of this buffer is controlled by Write Enable, ensuring that:

- Data reaches the write module only during valid write cycles
- The write driver remains isolated otherwise

Each bitline is connected to a dedicated tri-state buffer for readout. The enable of this buffer is tied to the inverse of the clocked Write Enable, serving as the Read Enable signal. This ensures:

- Readout occurs only when the array is in read mode

Shared modules:

- Non-overlapping clock generator
- 4-to-16 row decoder
- Clocked Write Enable / Read Enable logic

Per-column modules:

- Bitline precharge
- Bitline equalizer
- Column write driver
- Column read tri-state buffers
- Local SRAM cells (16 per column)

This division ensures efficient layout while maintaining modularity and clean design hierarchy. The full schematic for this design is shown in Figure 22. The decoder and clock generation, precharging, and write and read module sections of the design are included in Figures 23, 24, and 25, respectively.

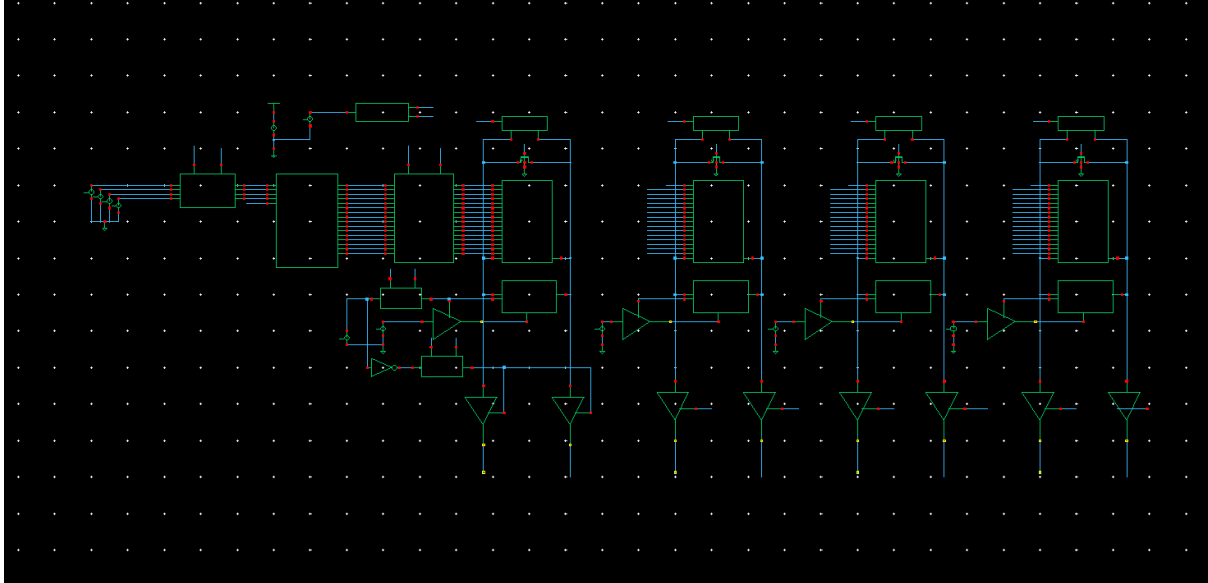


Figure 22. Fully synchronous, 64 bit 6T SRAM array Complete with peripheral circuitry.

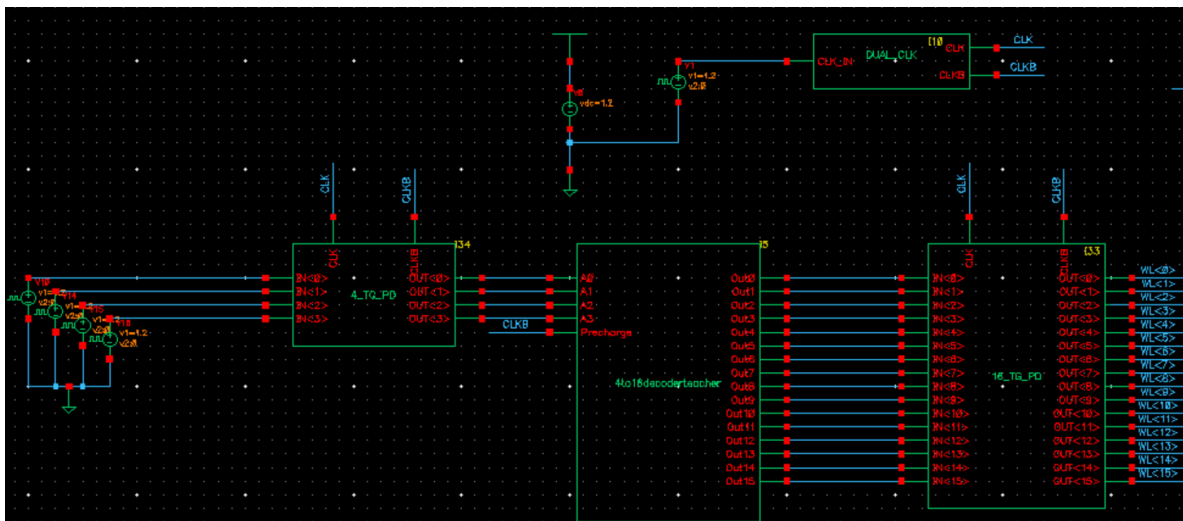


Figure 23. Non-overlapping clock generator. Clocked decoder input and output. Dynamic cells in decoder are precharged with CLKB.

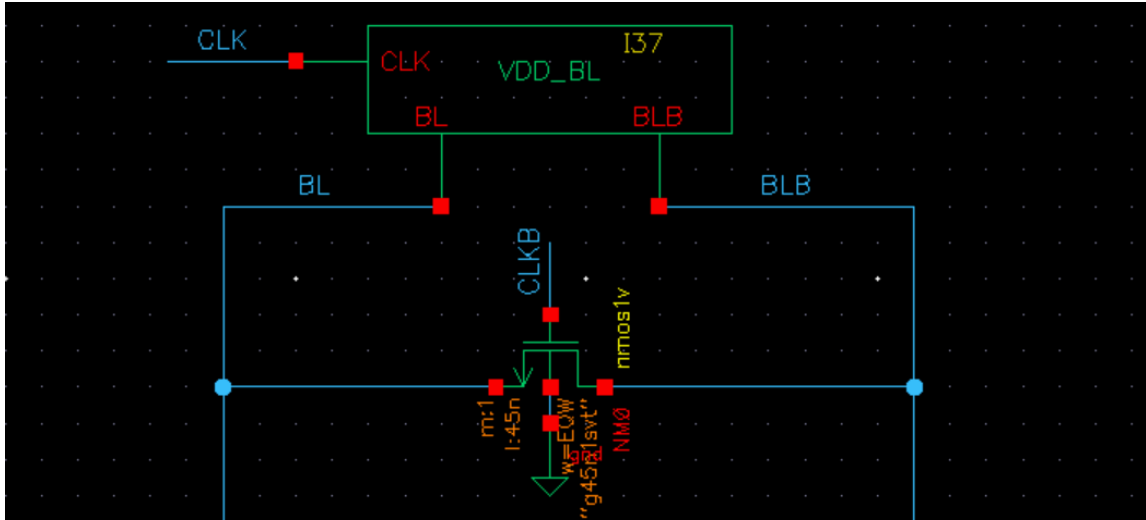


Figure 24. Bitline precharging circuit. Complete with clocked PMOS VDD precharge module and EQ NMOS transistor. All precharging performed during CLKB high.

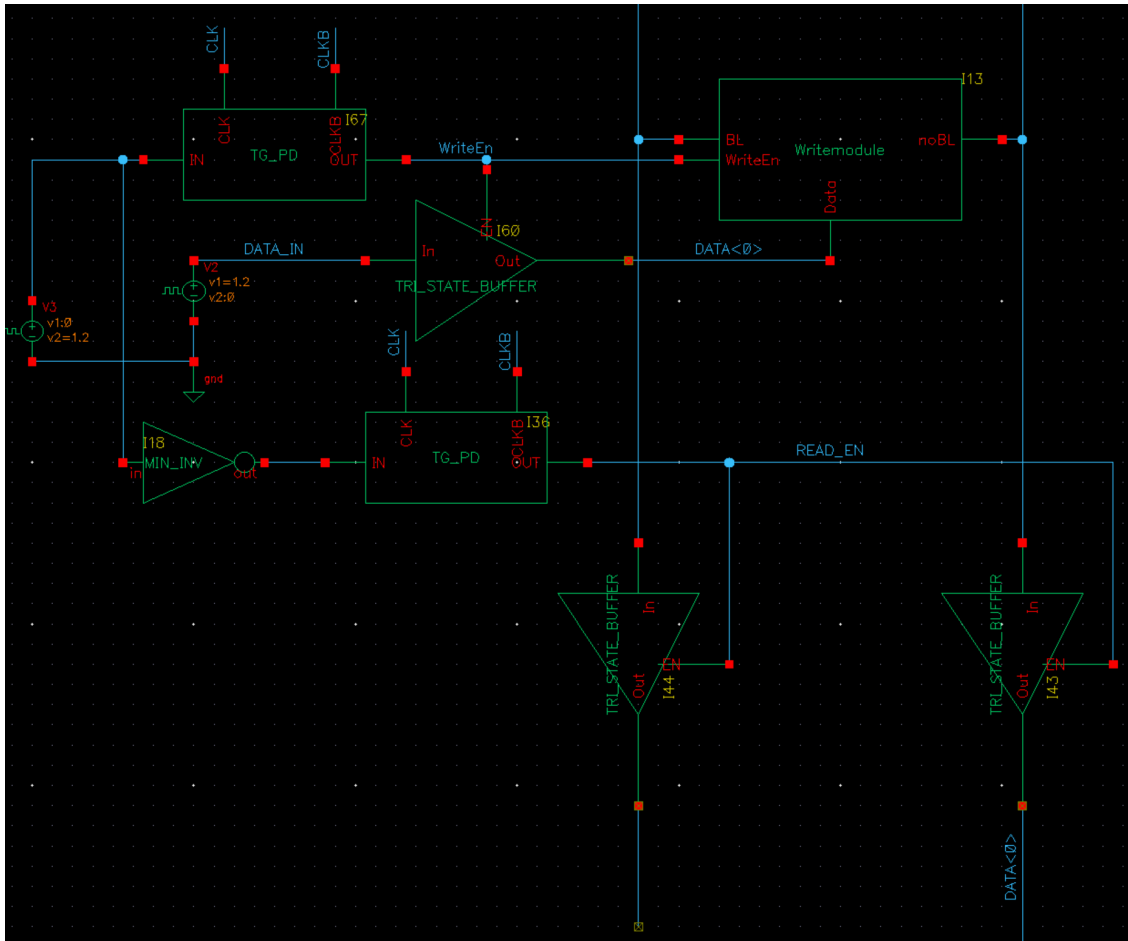


Figure 25. Read/Write enabling circuitry. Clocked Write Enable signal feeds into the write module. Input data buffered onto the shared input/output data bus. Write Enable signal inverted and clocked to create Read Enable.

The schematic in Figure 22 also serves as the test schematic for the entire cell. The following test sequence was applied to the cell to verify correct operation:

WL 0: W0 R0 R0 W1 R1 R1 W1 R1 R1 W0 R0 R0
 WL 15: W0 R0 R0 W1 R1 R1 W1 R1 R1 W0 R0 R0

Figure 26 includes the results from that simulation,

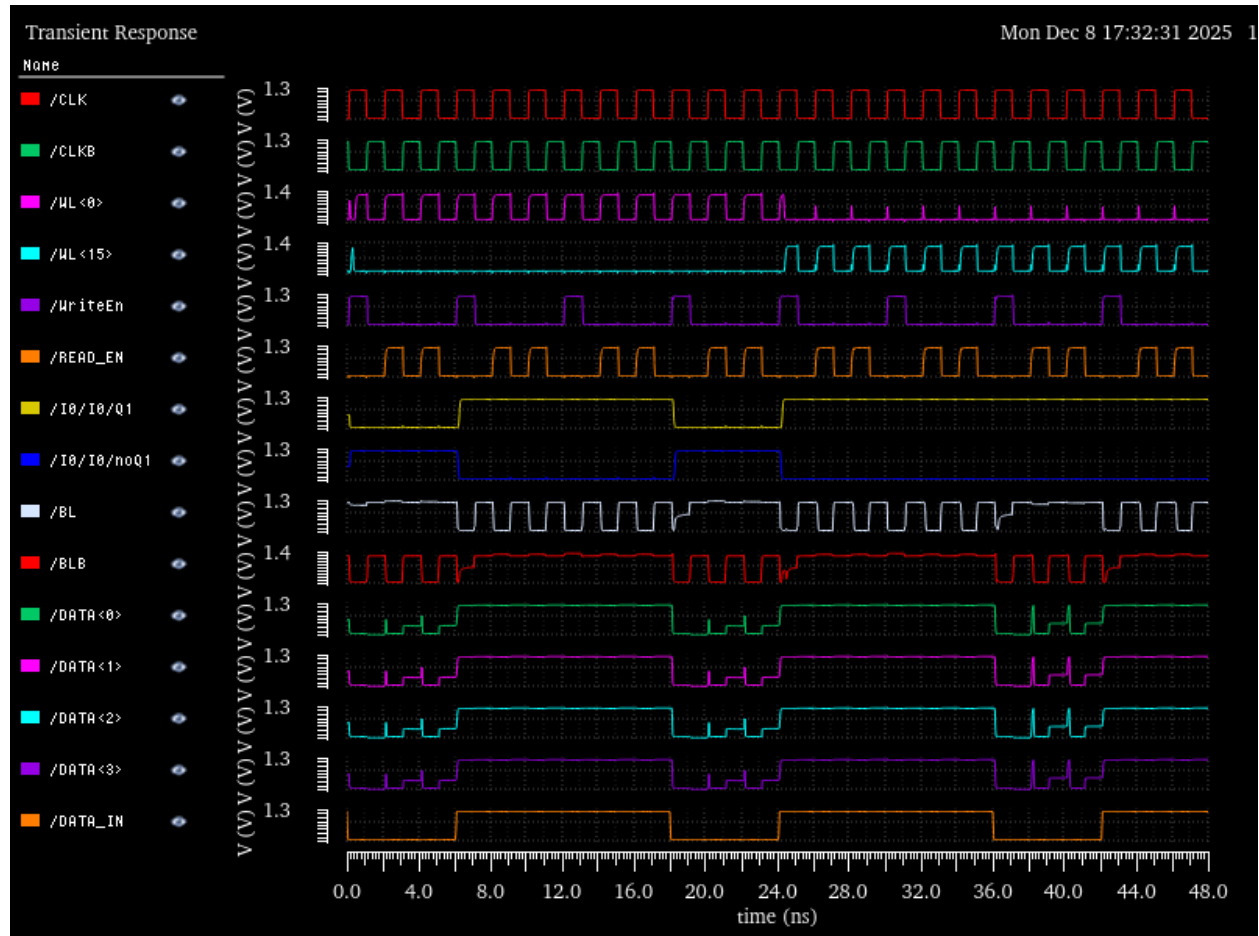


Figure 26. Test output waveforms for the full 64 bit SRAM array. Write and read operations are performed on WL 0 for the first half of the test and WL 15 for the second half.

The results of the simulation verify the key functions of the SRAM cells and the periphery circuitry.

- Only one word line is active during CLK high
- All word lines are forced low during CLKB high
- Bitlines are fully precharged and equalized each cycle
- Writes occur only when both the word line and clocked Write Enable are high
- Reads occur only when Read Enable is active

3. Optimized Design

The optimized design maintains the overall structure and hierarchy of the baseline architecture but introduces several key circuit-level improvements aimed at reducing the overall FOM. The primary refinements include replacing full-VDD bitline precharge with a half-VDD precharge network, integrating an operational transconductance amplifier (OTA) based differential sense amplifier, and adopting minimum-sized SRAM cell transistors enabled by the reduced bitline swing. The following subsections describe these enhancements and their verification results.

3.1 Half-VDD Bitline Precharge Network

In the optimized design, the baseline full-VDD bitline precharge is replaced with a mid-supply ($V_{DD}/2$) precharge scheme. This significantly reduces bitline capacitance charging energy and improves read access time by decreasing the initial bitline voltage difference the cell must overcome during a read.

The VDD/2 precharging module uses a diode-connected PMOS and NMOS stack, with both drains tied to the input of an inverter. The inverter stabilizes the midpoint and buffers the VDD/2 level at its output. This buffered half-VDD signal is then routed to two NMOS pass transistors whose gates are controlled by CLK_B.

When CLKB is high (precharge phase), both BL and BLB are pulled to $V_{DD}/2$ through these pass transistors. When CLKB is low (active phase), the pass devices turn off, isolating the bitlines for read/write operations.

This approach yields lower dynamic power, faster differential development during reads, and reduced risk of read disturb.

The VDD/2 precharging circuit is shown in Figure 27.

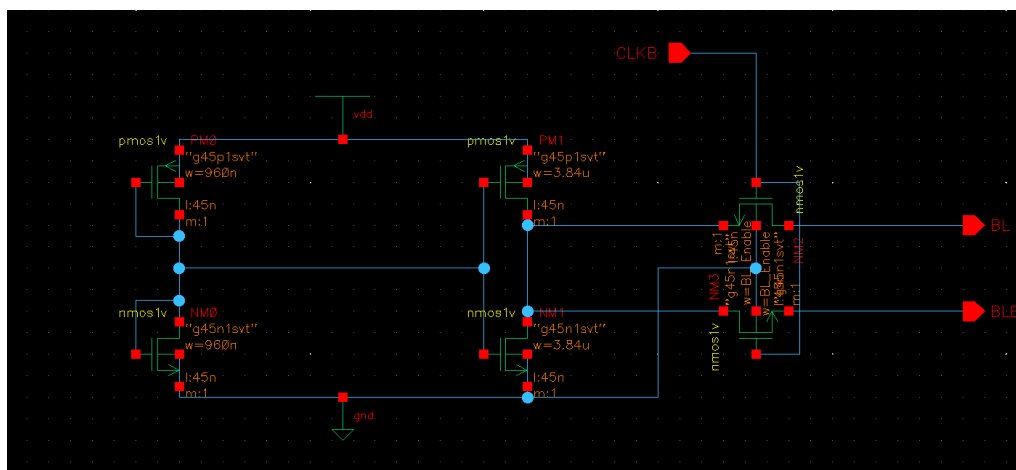


Figure 27. VDD/2 precharge circuit. Stacked diode connected PMOS/NMOS create a halfway

reference voltage, which the inverter holds at its output. Transmission gates pass along the signal to the bitlines.

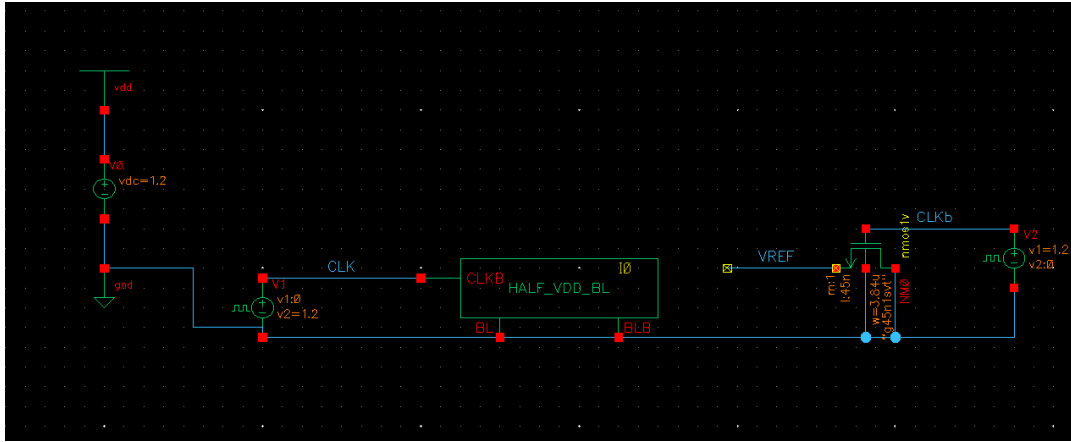


Figure 28. Test schematic for VDD/2 precharge circuit. The output is a reference voltage.

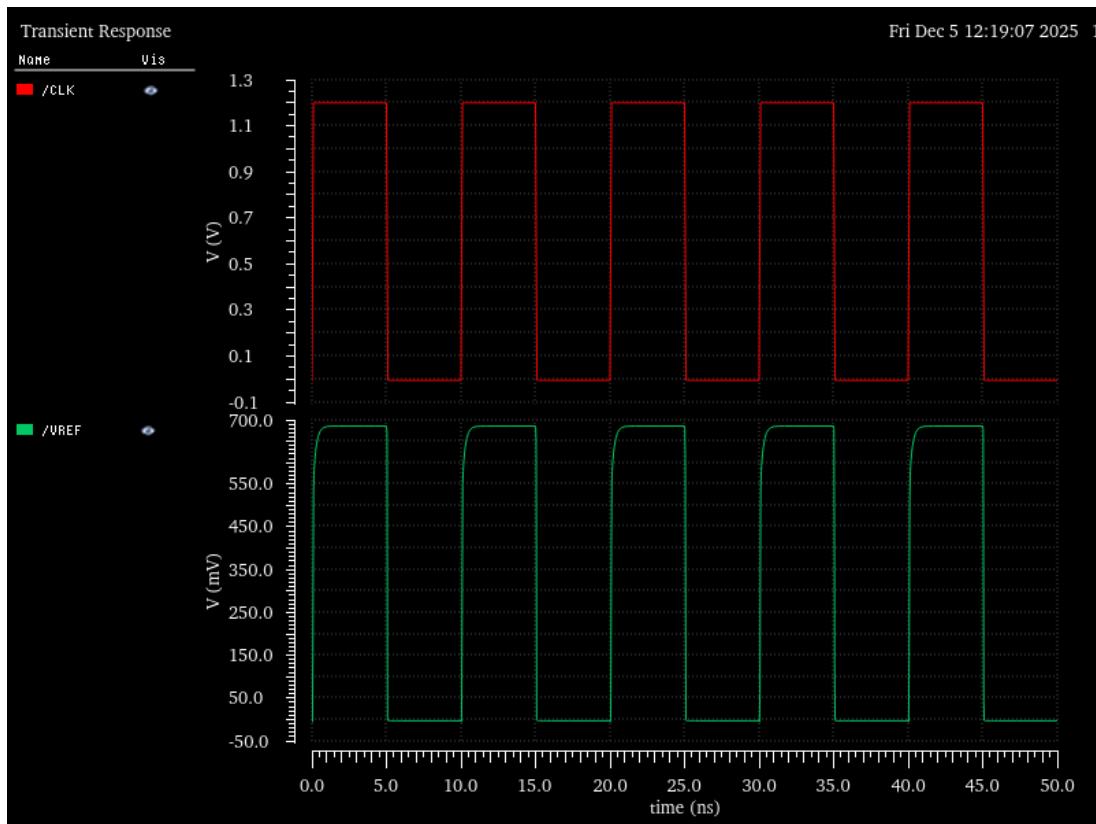


Figure 29. The circuit outputs a steady reference voltage of 0.685V when the input clock is high.

3.2 Differential Sense Amplifier

To replace the baseline tri-state output buffers and improve read sensitivity, the optimized design incorporates an OTA-style differential sense amplifier. This sense amplifier compares the small voltage difference created on BL and BLB during a read and generates a full-swing digital output.

The topology consists of:

- A differential NMOS input pair connected directly to BL and BLB
- PMOS active load devices forming a differential-to-single-ended conversion
- A tail NMOS current source whose gate is driven by the Read Enable signal

The sense amplifier remains inactive when Read Enable is low, consuming no static power. When Read Enable is asserted, the tail current source biases the OTA, amplifying the small BL-BLB difference into a full logic-level swing at the output.

Figure 30 includes the sense amplifier topology.

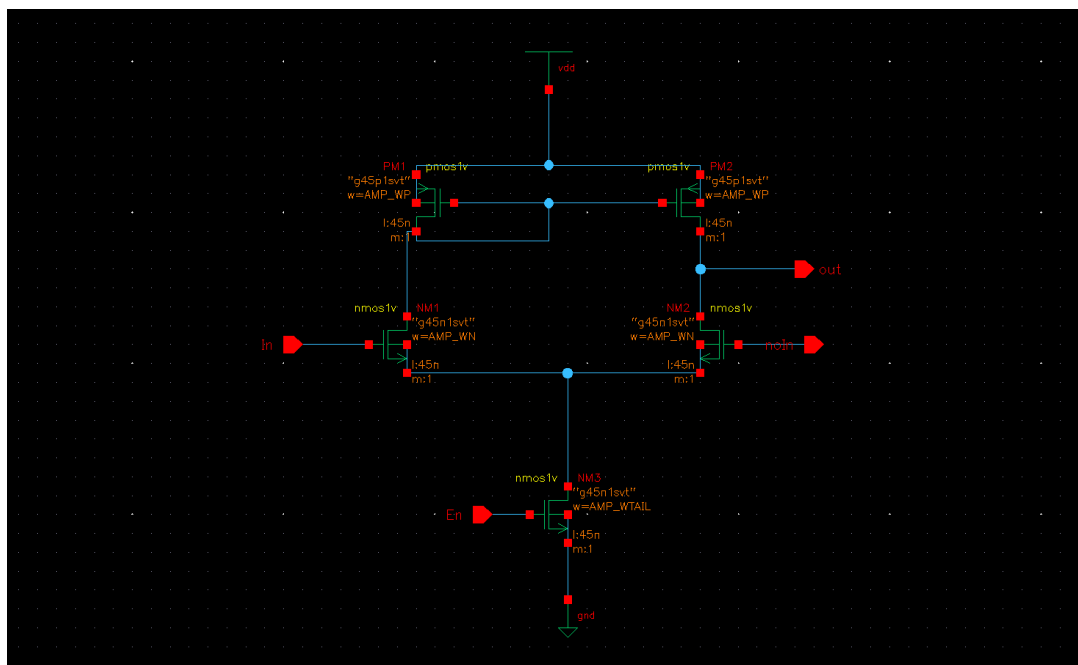


Figure 30. The sense amplifier is an operational transconductance amplifier, providing a single ended output from a differential input.

3.3 Minimum-Sized 6T SRAM Cell

Because the bitlines now begin at $VDD/2$ instead of VDD , the read disturb margin is improved: the cell does not have to hold its internal node against a full-swing bitline voltage. This allows the memory cell to be constructed with minimum-sized transistors.

Benefits of VDD/2 Precharge:

- During reads, the pull-down NMOS transistors in the cell experience a smaller voltage difference between their internal nodes and the bitlines, which reduces the risk of unintentionally flipping the stored value.
- Writeability is maintained because the write driver still imposes a full differential voltage.
- Leakage and dynamic power are reduced due to smaller device sizes.

3.4 Full 64-Bit Optimized Array

The full 64-bit array retains the overall four column by sixteen cell structure. However, the optimized array integrates the optimized peripherals.

Architectural Changes

- Half-VDD precharge replaces the baseline full-VDD bitline precharge in all columns.
- Sense amplifiers replace the column-level output tri-state buffers, providing faster and more robust readout.
- All 6T cells in the array use minimum-sized devices.
- Global signals (CLK/CLKB, decoder outputs, Write Enable, Read Enable) remain as in the baseline, maintaining compatibility with the optimized per-column circuitry.
- As before, the decoder, clock generator, and control logic are shared across the array, while each column maintains independent precharge, equalization, and sensing circuitry.

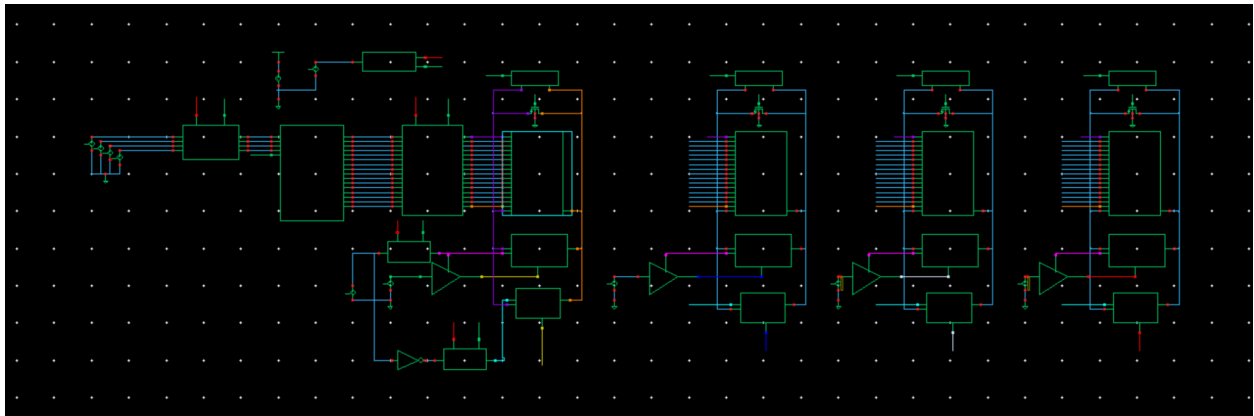


Figure 31. Optimized 64 bit 6T SRAM cell. Includes VDD/2 precharge, sense amp, and min size sram cells.

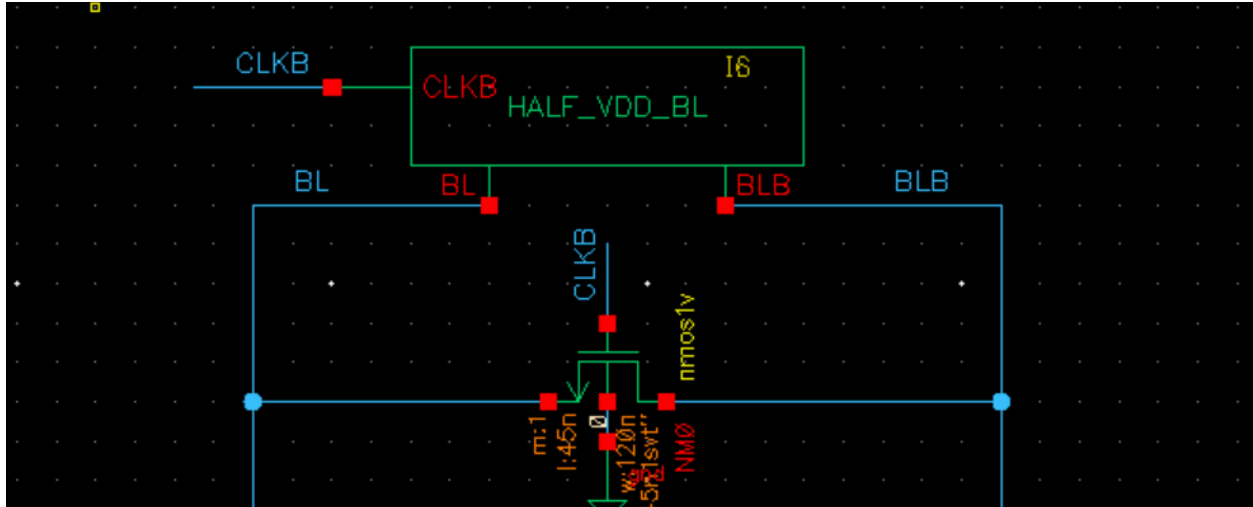


Figure 32. VDD/2 precharge module connected to each bitline and controlled by CLKB.

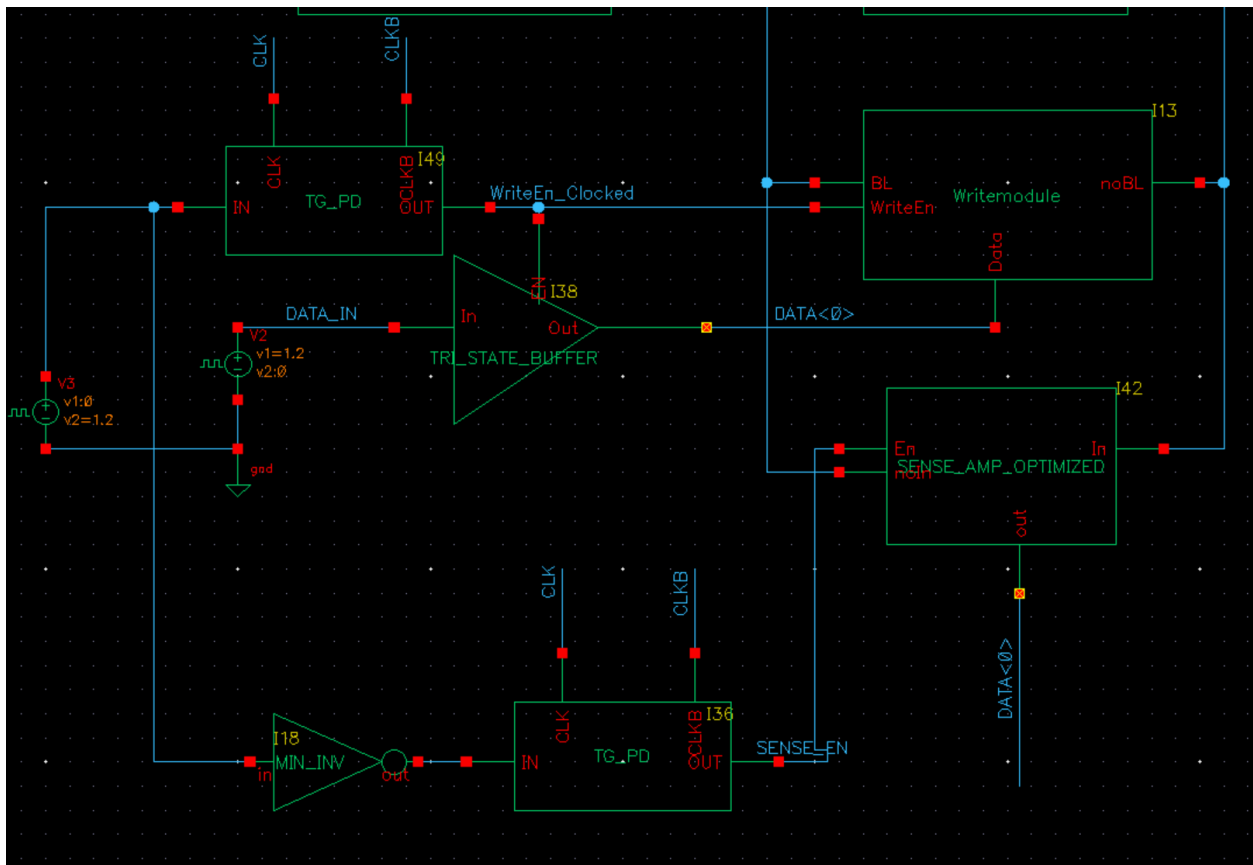


Figure 33. Read/Write circuitry for the optimized array. Replaces the tri state inverters attached to the bitline with the sense amp.

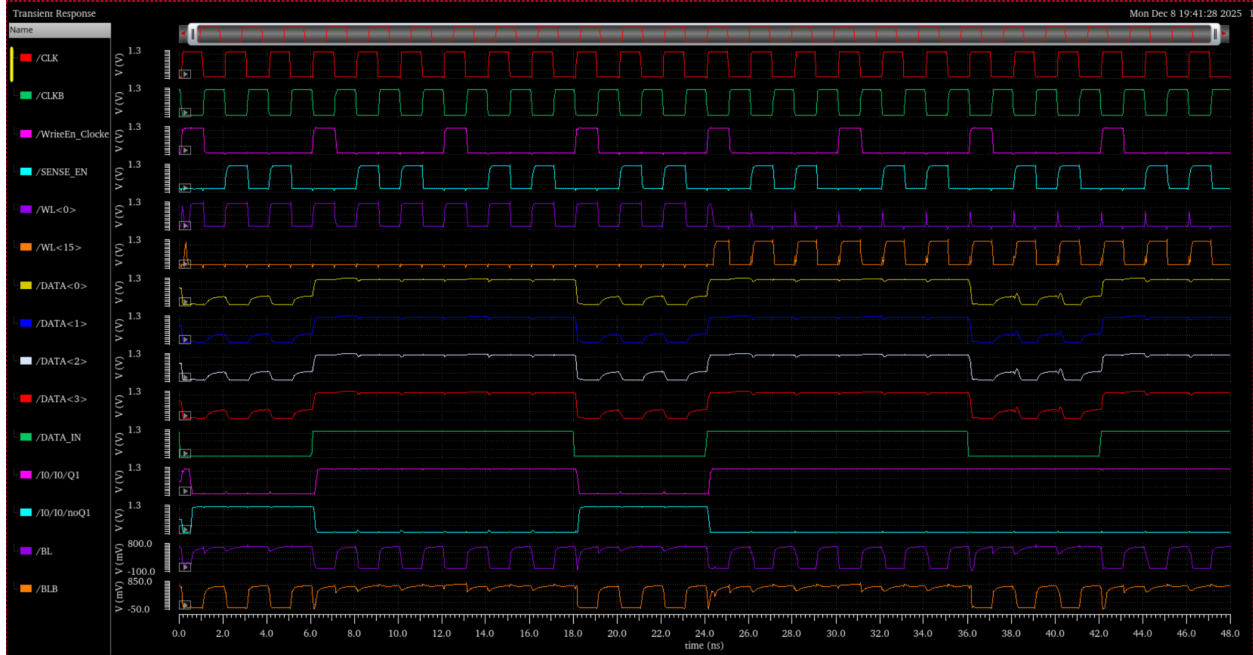


Figure 34. Test output waveforms for optimized array. Bitlines are charged to near $V_{DD}/2$.

Array-level simulations confirm:

- Reliable generation of $V_{DD}/2$ on all bitlines during precharge
- Clean transition from precharge to evaluation through the non-overlapping clock waveforms
- Correct wordline selection via the clocked decoder interface
- Fast and accurate sensing from the OTA-based sense amplifiers
- Successful writes using the same write driver architecture as in the baseline

4 Performance Comparison

This section presents a quantitative comparison of the FOM between the baseline 6T SRAM cell and the optimized design. As shown in Eqn. 1, FOM is broken down into area, delay, and power. Write and read access delays will be determined for each design, and from that, the maximum switching frequency will be determined. With this frequency determined, the power draw of each circuit will be measured.

The methodology for extracting read access time and write access time is described first, followed by measured results for each design and a consolidated comparison.

4.1 Methodology for Measuring Access Times

The read access time is defined as the delay between the activation of the wordline (WL) and the measurement of the cell's data contents. This will be measured from 50% of the wordline swing to 50% of the data bus swing.

The write access time is defined as the time between the activation of the wordline and the moment when the storage node Q successfully flips to the intended value. This will be measured from 50% of the wordline to 50% of the storage node swing.

4.2 Methodology for Measuring Power Draw

The power consumption of the memory cell is defined as the average energy consumed per second during repeated write operations across the entire array. Specifically, the array is written with all 1s and then all 0s for four full cycles each, with every write performed at the maximum allowable switching frequency of the array.

Determining this maximum switching frequency begins with assessing the worst-case access time delay, but unlike combinational logic, the maximum frequency of an SRAM array is not simply the inverse of this delay. SRAM write operations require additional setup and hold time to ensure that the new data value is fully established and retained at the storage node. If these timing requirements are not met, the written value may be unstable, corrupted, or completely overwritten. As a result, both the access delay and the data retention timing constraints must be considered when identifying the highest safe operating frequency for power measurements.

To find this maximum switching frequency, or minimum clock period, a small parametric sweep will be performed, starting with the largest read access time. The largest frequency that exhibits proper operation at the end of the clock cycle will be used.

4.3 Baseline SRAM Cell Performance

The read and write access times of the baseline SRAM cell were measured first. To characterize the high-to-low write access time, a logic '1' was written into a cell initially storing a '0,' and the resulting delay was recorded. The low-to-high write access time was measured using the opposite transition.

Figures 35 and 36 depict write operations from low to high and high to low, respectively.

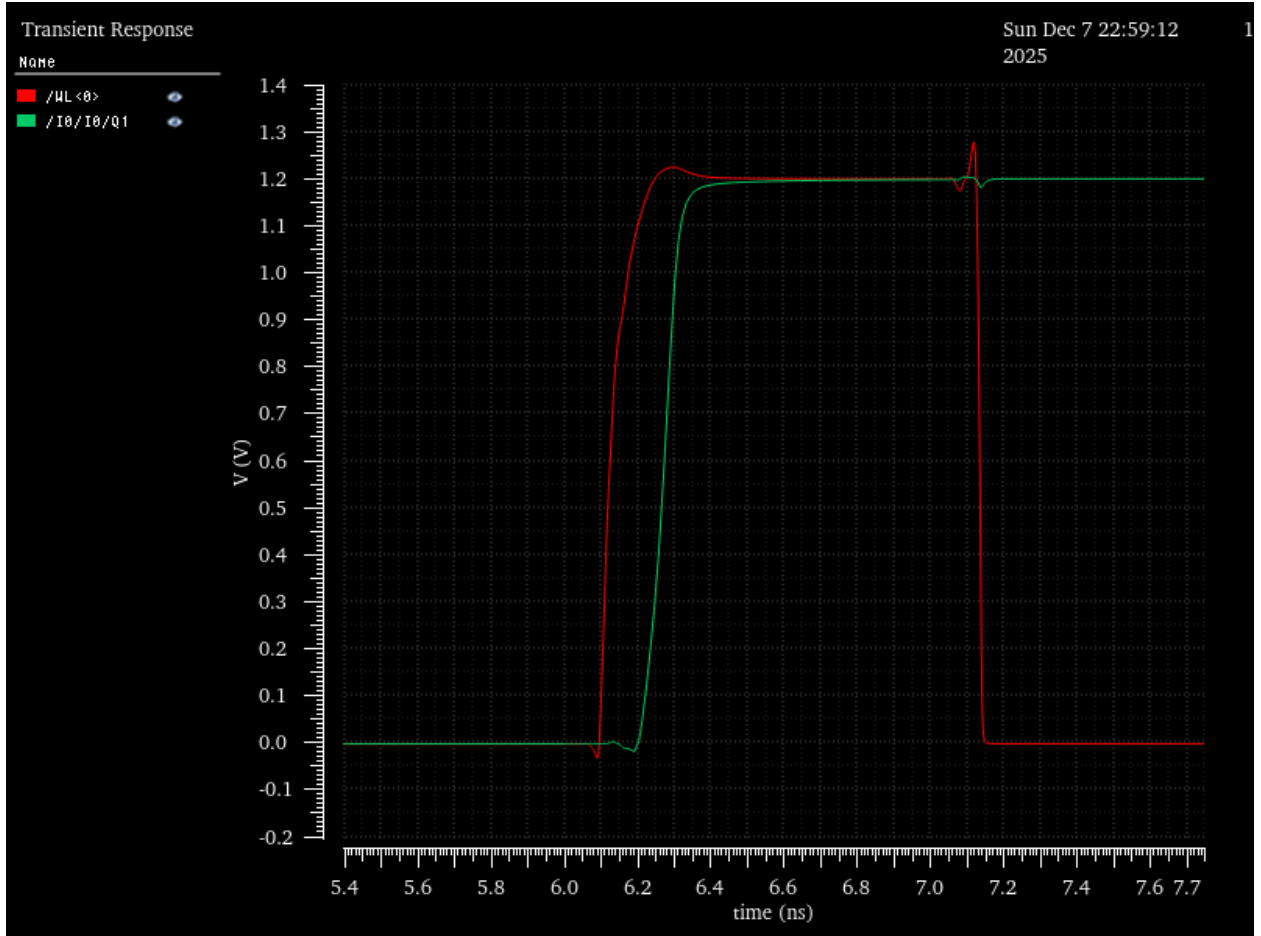


Figure 35. Measurement of the write access time from low to high in the baseline SRAM array

$$WA_{LH} = 6.2727ns - 6.1243ns = 148ps$$

Eqn. 2

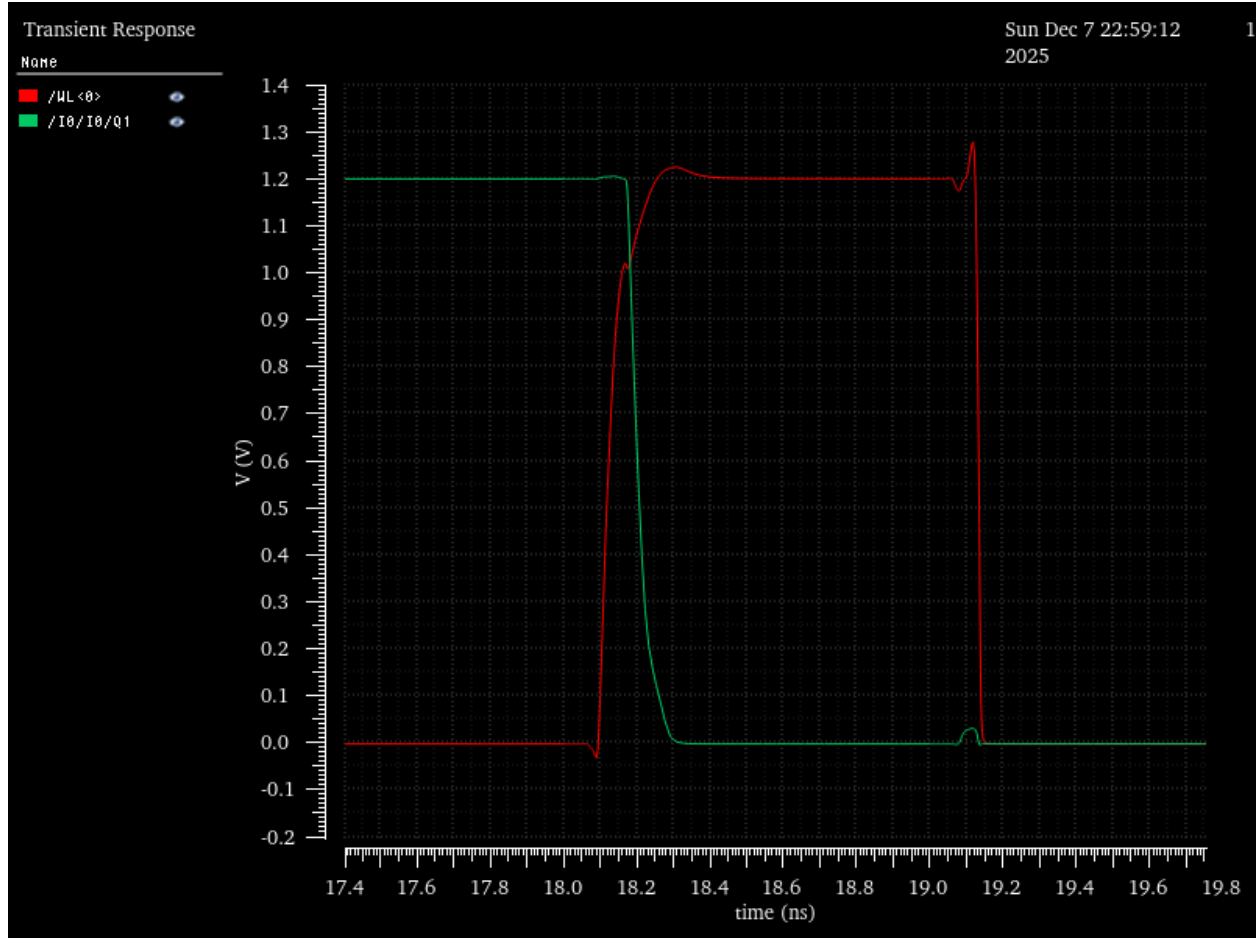


Figure 36. Measurement of the write access time from high to low in the baseline SRAM array

$$WA_{HL} = 18.2020ns - 18.1236ns = 78.4 ps$$

Eqn. 3

Measuring the read access times required an additional step. Because the data bus is shared between the array's input and output, its value immediately after a write inherently matches the stored cell value. As a result, simply enabling the read operation does not produce a measurable transition at the data node. To emulate the worst-case read scenario, the data bus was intentionally forced to the opposite logic value before initiating the read. This ensures a meaningful transition at the output and yields a more accurate measurement of the worst-case read access delay.

Figures 35 and 36 depict the additional circuitry employed to achieve this data forcing for the high to low and low to high transitions, respectively.

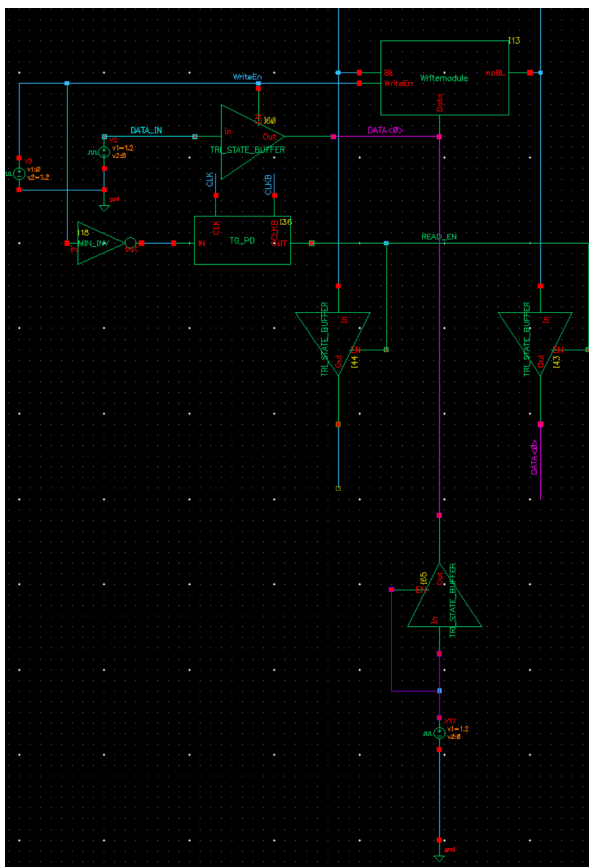


Figure 35. Peripheral circuitry to force a '1' to the data bus

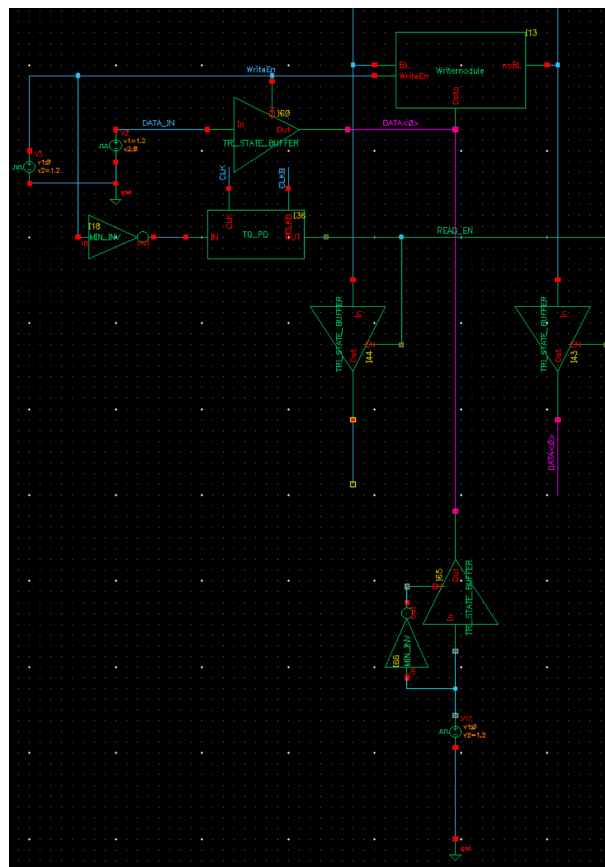


Figure 36. Peripheral circuitry to force a '0' to the data bus

Figures 37 and 38 show the resulting read access delay measurements.

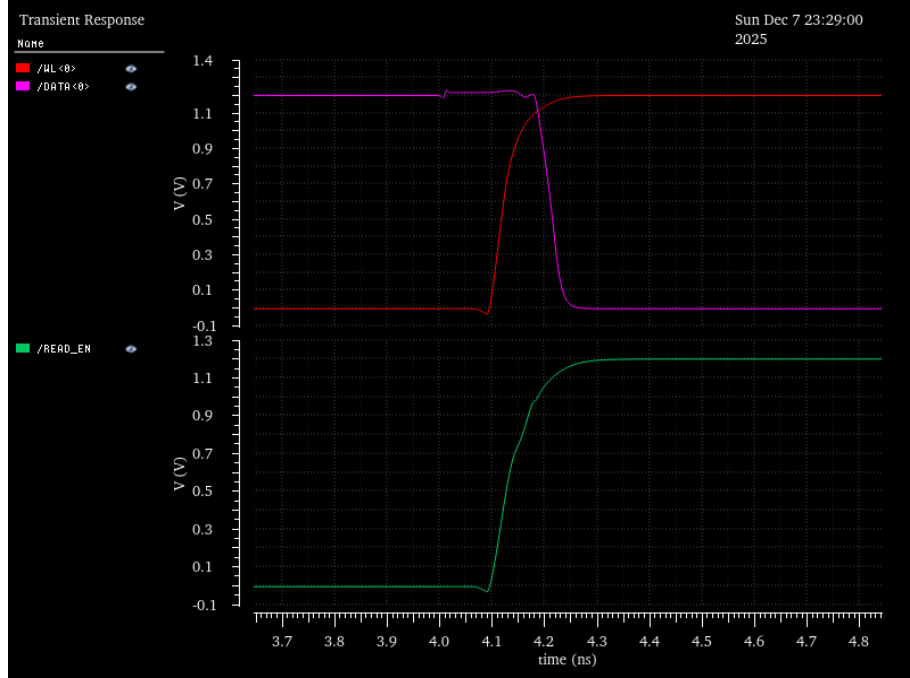


Figure 37. Measurement of the read access time for a '0' in the baseline SRAM array

$$RA_0 = 4.2110ns - 4.1215ns = 89.5 ps$$

Eqn. 4

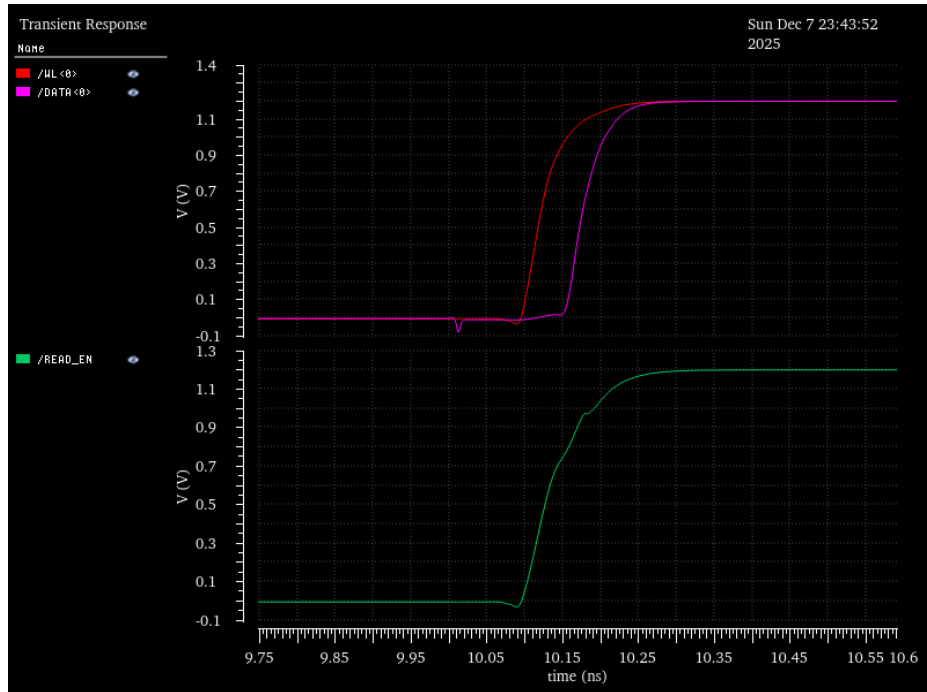


Figure 38. Measurement of the read access time for a '1' in the baseline SRAM array

$$RA_1 = 10.1763ns - 10.1216ns = 54.7 ps$$

Eqn. 5

The four delay times are compiled in Table 1.

Table 1. Access time delays for the baseline memory array design.

W/R	Write Access		Read Access	
Direction	High-Low	Low-High	Low	High
Delay (ps)	78.4	148	89.5	54.7

The longest access delay in the baseline design is 148 ps.

With this minimum delay measured, a parametric sweep was set up to determine the minimum possible delay.

Clock periods from 150ps to 400ps were swept. 11 points were used.

Starting from the shortest simulated clock period, the storage node Q was compared against the data input. As shown in Figure 39, at very short periods the storage node follows the rising transition of the data input but fails to return low when the data input switches low again, indicating insufficient time for a full write operation. The first clock period at which the storage node correctly followed both the rising and falling transitions of the data input was 250 ps.

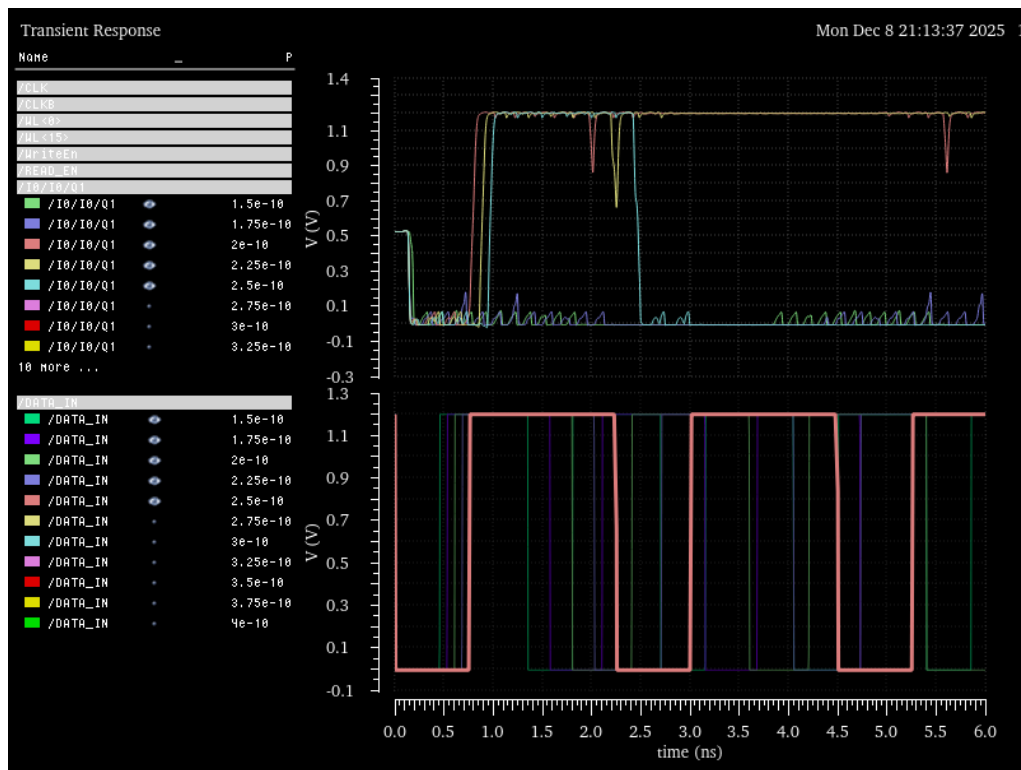


Figure 39. A parametric sweep of clock periods for the baseline design. Since the period of 250 ps is the first whose data storage node Q follows the input data, it is further investigated as the potential minimum period.

To verify that a 250 ps clock period was sufficient for correct end-of-cycle behavior, the output data line was compared to the read-enable signal, as shown in Figure 40. Although the data line temporarily deviates from the correct value during the cycle—due to shared I/O and internal bitline dynamics—it returns to the correct state by the end of the clock period. Since the output is valid at the clock boundary, the array is considered to operate correctly at a 250 ps minimum clock period.

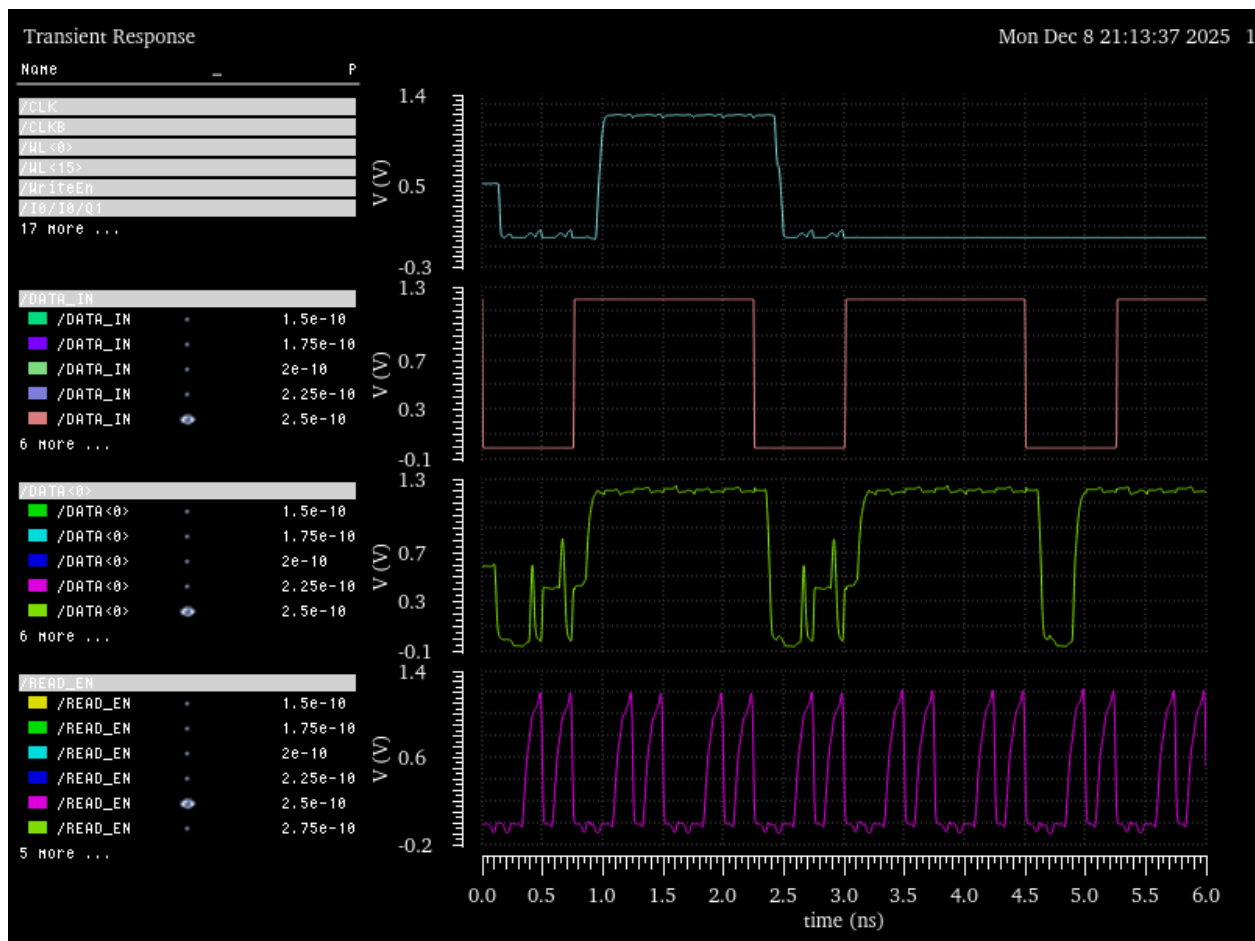


Figure 40. Upon comparing the data bus response to the read enable signal, it is clear that the data does return to the correct output at the very end of the clock cycle, signifying that this period yields correct operation.

With the minimum period established, the average power draw can be measured. Figure 41 shows the supply current during successive write operations. Z

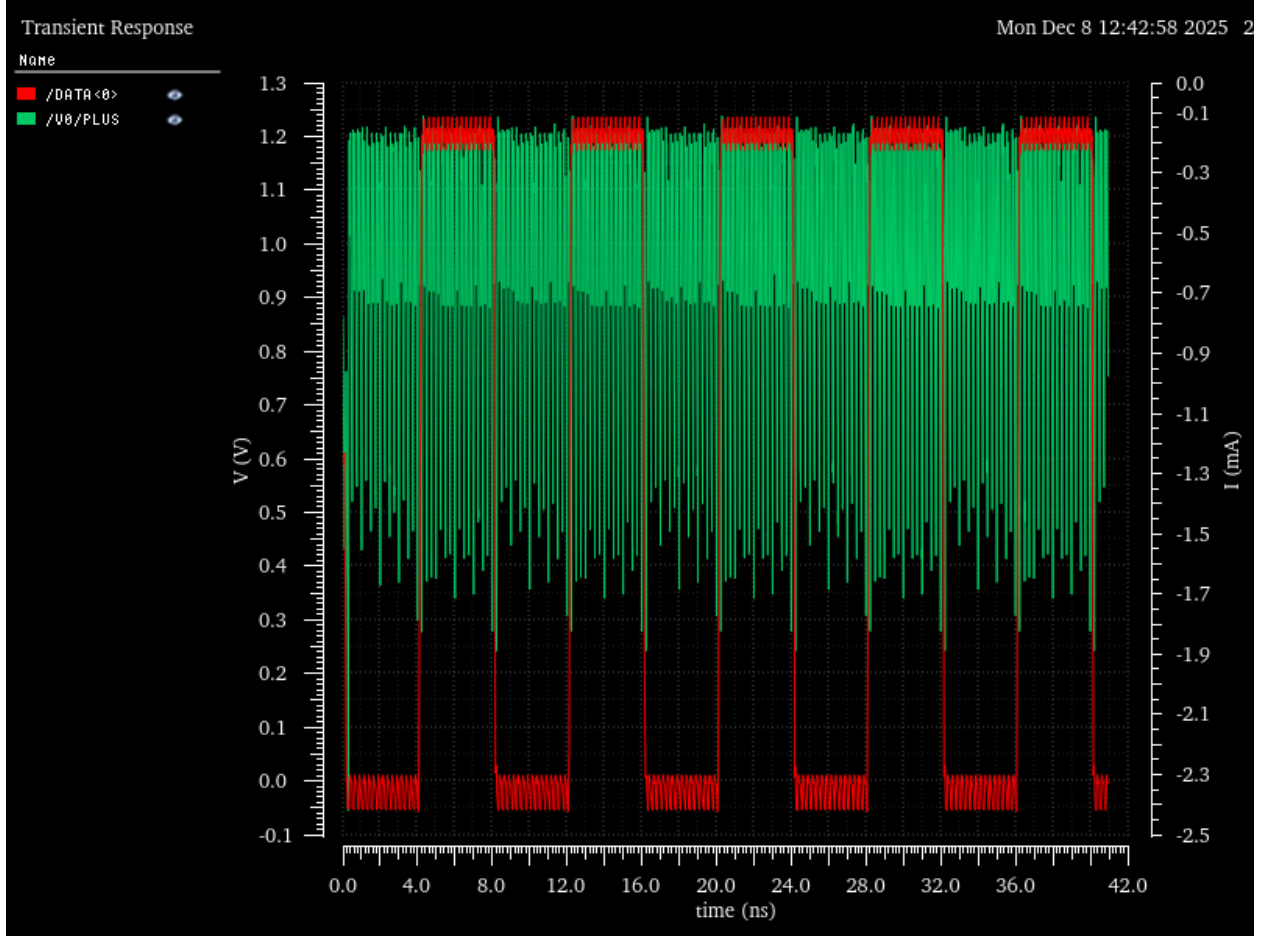


Figure 41. Current draw during the successive write operations performed on the sram cell.

Eqn. 6 summarizes the method used to compute the array's average power consumption.

$$P = \frac{\int_{ti}^{tf} |I| dt \times V}{tf - ti}$$

Eqn. 6

Using Eqn. 6, the power draw of the baseline array was determined to be 713 uW.

The area of the array is assumed to be the total area of the SRAM cells. Eqn. 7 determines the area of the baseline array.

$$Area = 16 \times 4 \times [(120nm \times 2) + (240nm \times 2) + (960nm \times 2)] \times 45nm = 760320nm^2 = 7.6\mu m^2$$

Eqn. 7

Using Eqn. 1, the FOM for the baseline array can be determined:

$$FOM = 7.6\mu m^2 \times .713mW \times 250ps = 1355\mu m^2 mWps$$

Eqn. 8

4.4 Optimized SRAM Cell Performance

The read and write access times of the optimized SRAM cell were measured first. To characterize the high-to-low write access time, a logic '1' was written into a cell initially storing a '0,' and the resulting delay was recorded. The low-to-high write access time was measured using the opposite transition.

Figures 42 and 43 depict write operations from low to high and high to low, respectively.

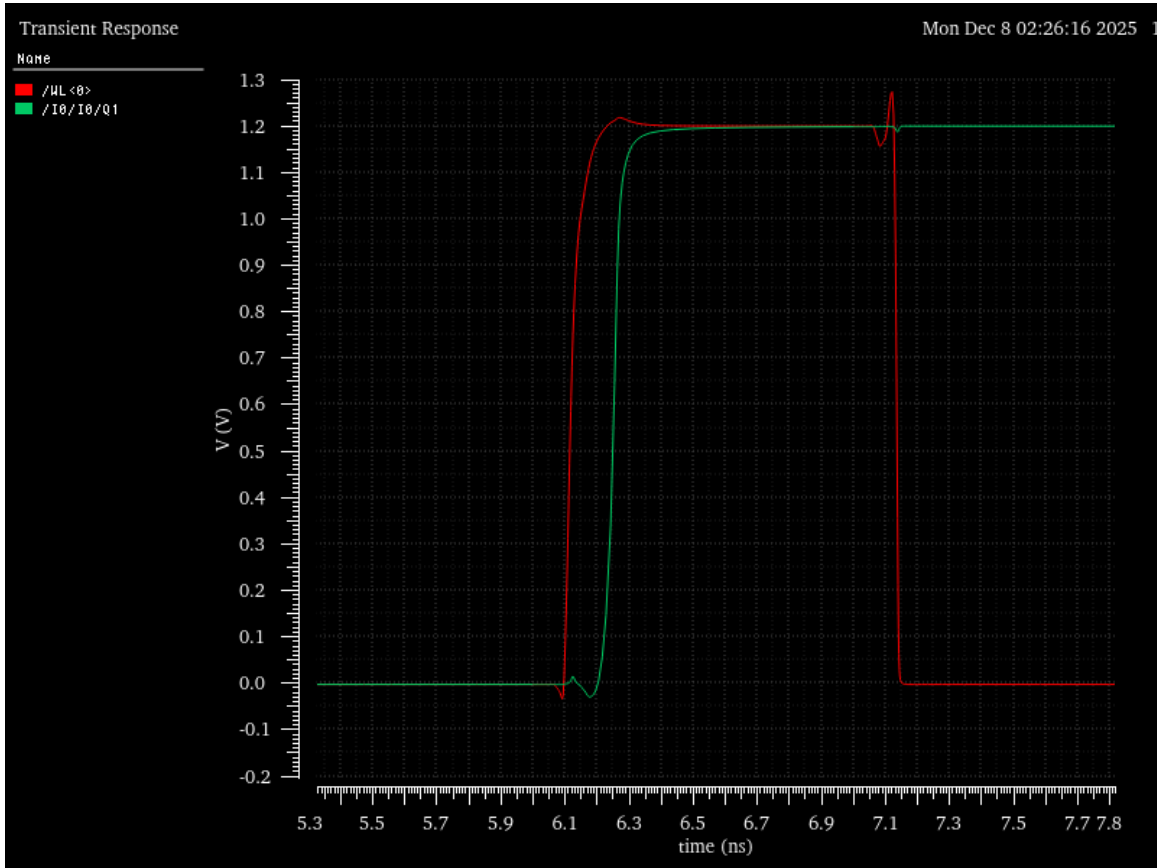


Figure 42. Measurement of the write access time from low to high in the optimized SRAM array

$$WA_{LH} = 6.2524ns - 6.1180ns = 134.4ps$$

Eqn. 9

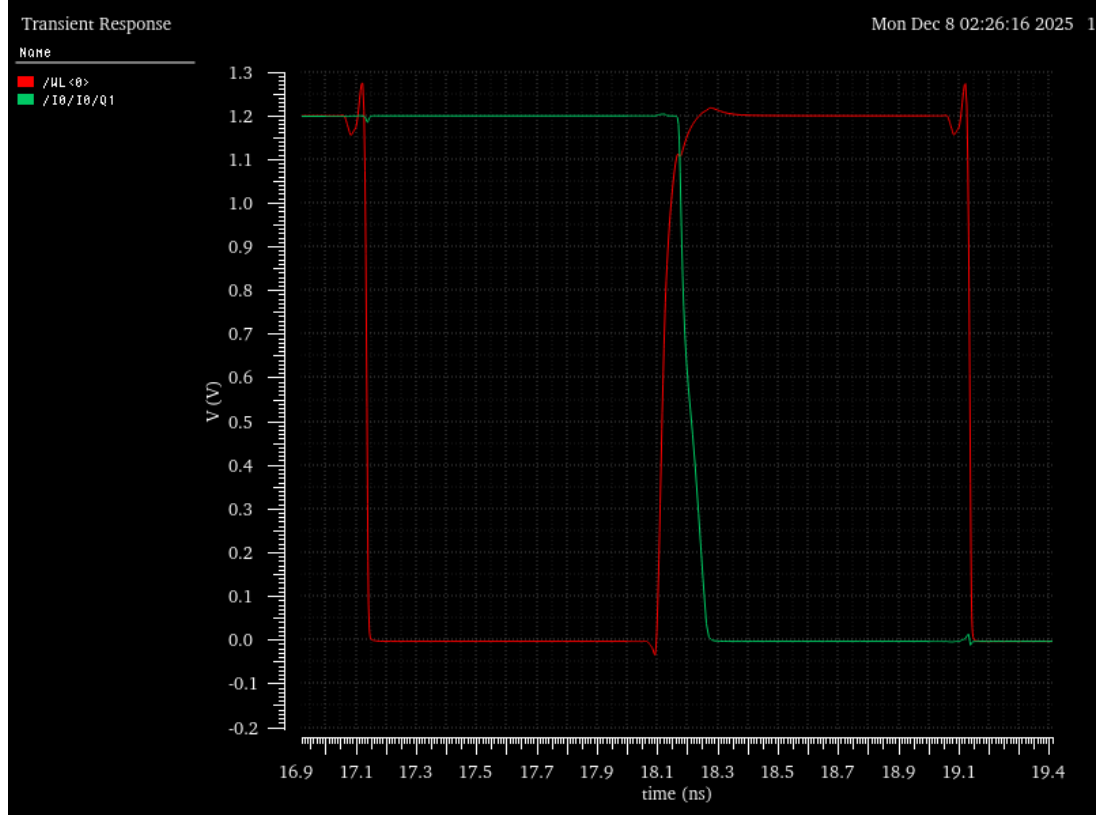


Figure 43. Measurement of the write access time from high to low in the optimized SRAM array

$$WA_{HL} = 18.222ns - 18.1180ns = 81.7 ps$$

Eqn. 10

Measuring the read access times required an additional step. Because the data bus is shared between the array's input and output, its value immediately after a write inherently matches the stored cell value. As a result, simply enabling the read operation does not produce a measurable transition at the data node. To emulate the worst-case read scenario, the data bus was intentionally forced to the opposite logic value before initiating the read. This ensures a meaningful transition at the output and yields a more accurate measurement of the worst-case read access delay.

Figures 44 and 45 depict the additional circuitry employed to achieve this data forcing for the high to low and low to high transitions, respectively.

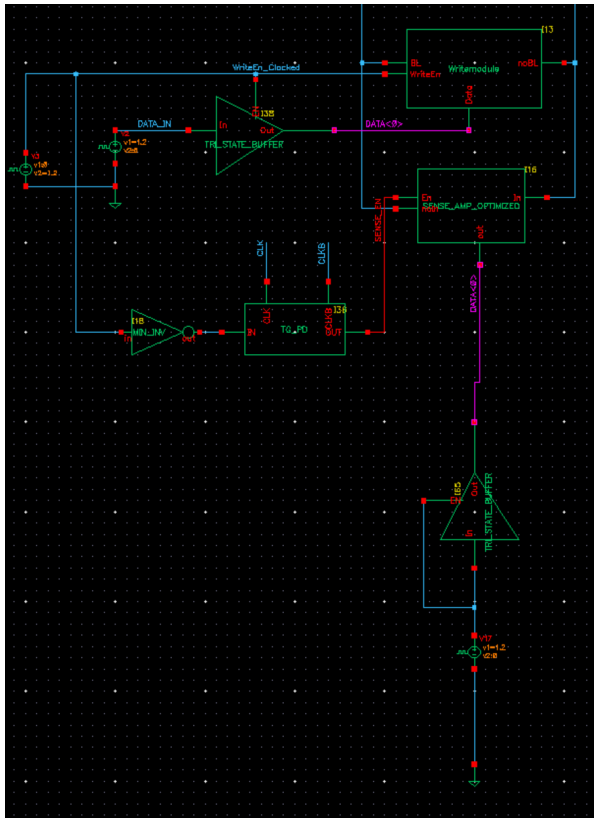


Figure 44. Peripheral circuitry to force a ‘1’ to the data bus

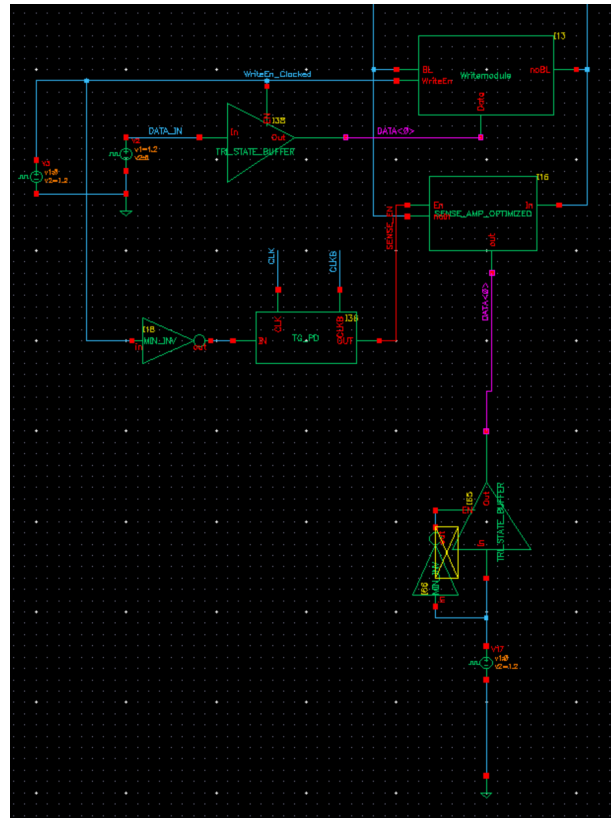


Figure 45. Peripheral circuitry to force a '0' to the data bus

Figures 46 and 47 show the resulting read access delay measurements.

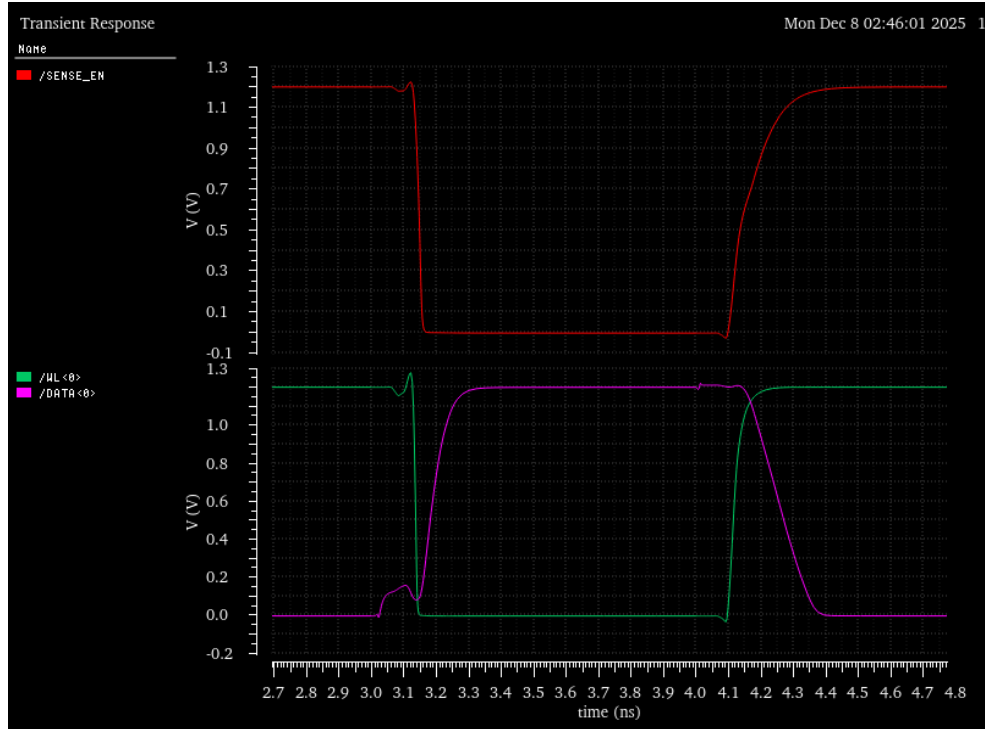


Figure 46. Measurement of the read access time for a '0' in the optimized SRAM array

$$RA_0 = 3.1879ns - 3.1352ns = 52.7 ps$$

Eqn. 11

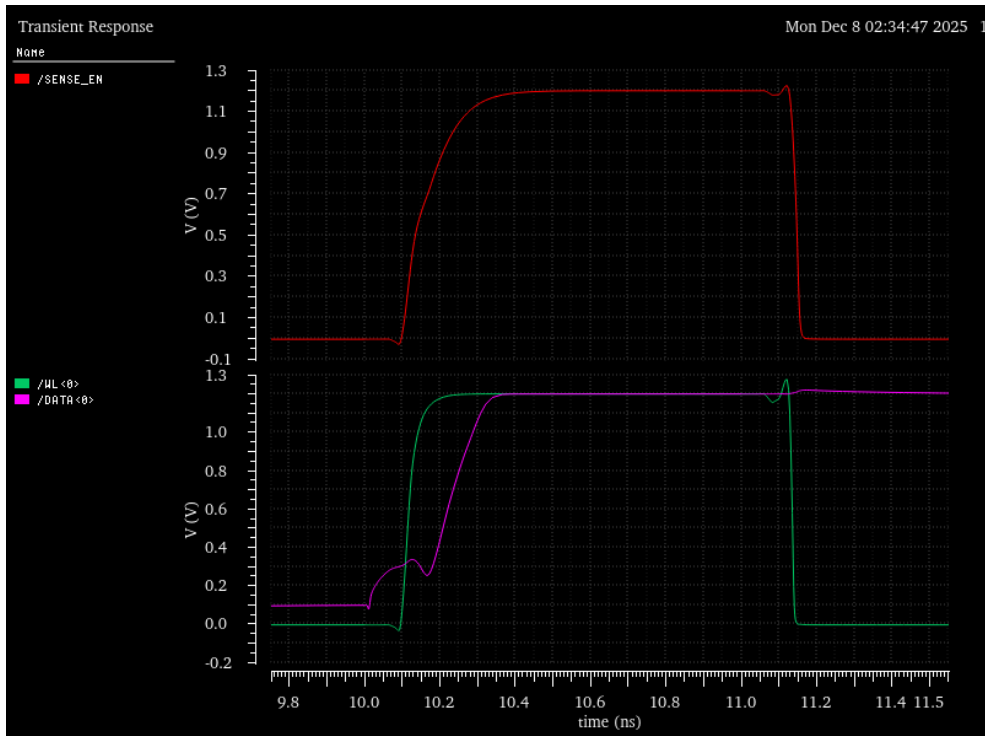


Figure 47. Measurement of the read access time for a '1' in the optimized SRAM array

$$RA_1 = 10.2393ns - 10.1178ns = 121.5 ps$$

Eqn. 12

The four delay times are compiled in Table 2.

Table 2. Access time delays for the optimized memory array design.

W/R	Write Access		Read Access	
Direction	High-Low	Low-High	Low	High
Delay (ps)	81.7	131.4	57.2	121.5

The longest access delay in the optimized design is 131.4 ps.

With this minimum delay measured, a parametric sweep was set up to determine the minimum possible delay.

Clock periods from 150ps to 300ps were swept. 11 points were used.

Starting from the shortest simulated clock period, the storage node Q was compared against the data input. As shown in Figure 48, at very short periods the storage node follows the rising transition of the data input but fails to return low when the data input switches low again, indicating insufficient time for a full write operation. The first clock period at which the storage node correctly followed both the rising and falling transitions of the data input was 255 ps.

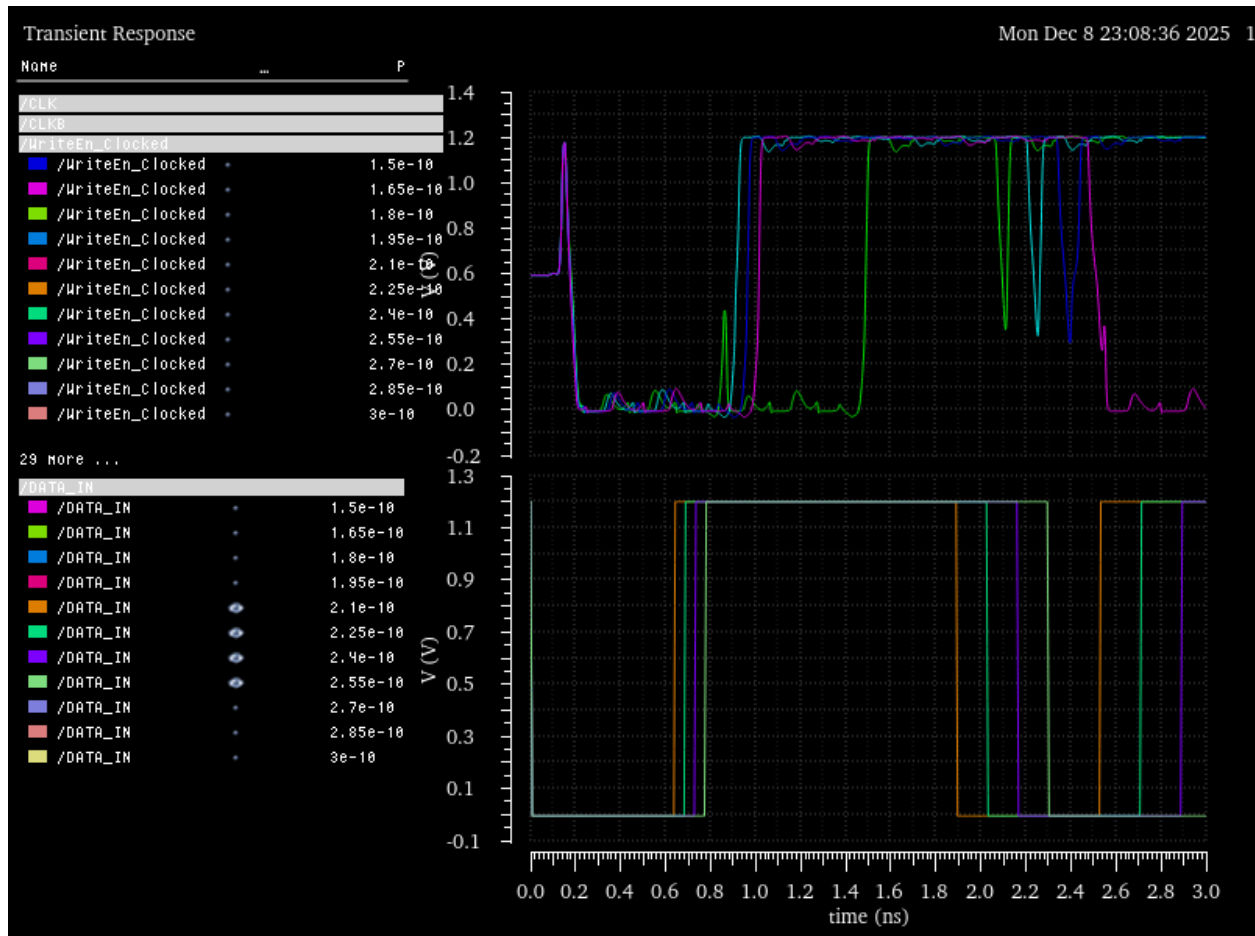


Figure 48. A parametric sweep of clock periods for the optimized design. Since the period of 255 ps is the first whose data storage node Q follows the input data, it is further investigated as the potential minimum period.

To verify that a 255 ps clock period was sufficient for correct end-of-cycle behavior, the output data line was compared to the read-enable signal, as shown in Figure 49. Although the data line temporarily deviates from the correct value during the cycle—due to shared I/O and internal bitline dynamics—it returns to the correct state by the end of the clock period. Since the output is valid at the clock boundary, the array is considered to operate correctly at a 255 ps minimum clock period.

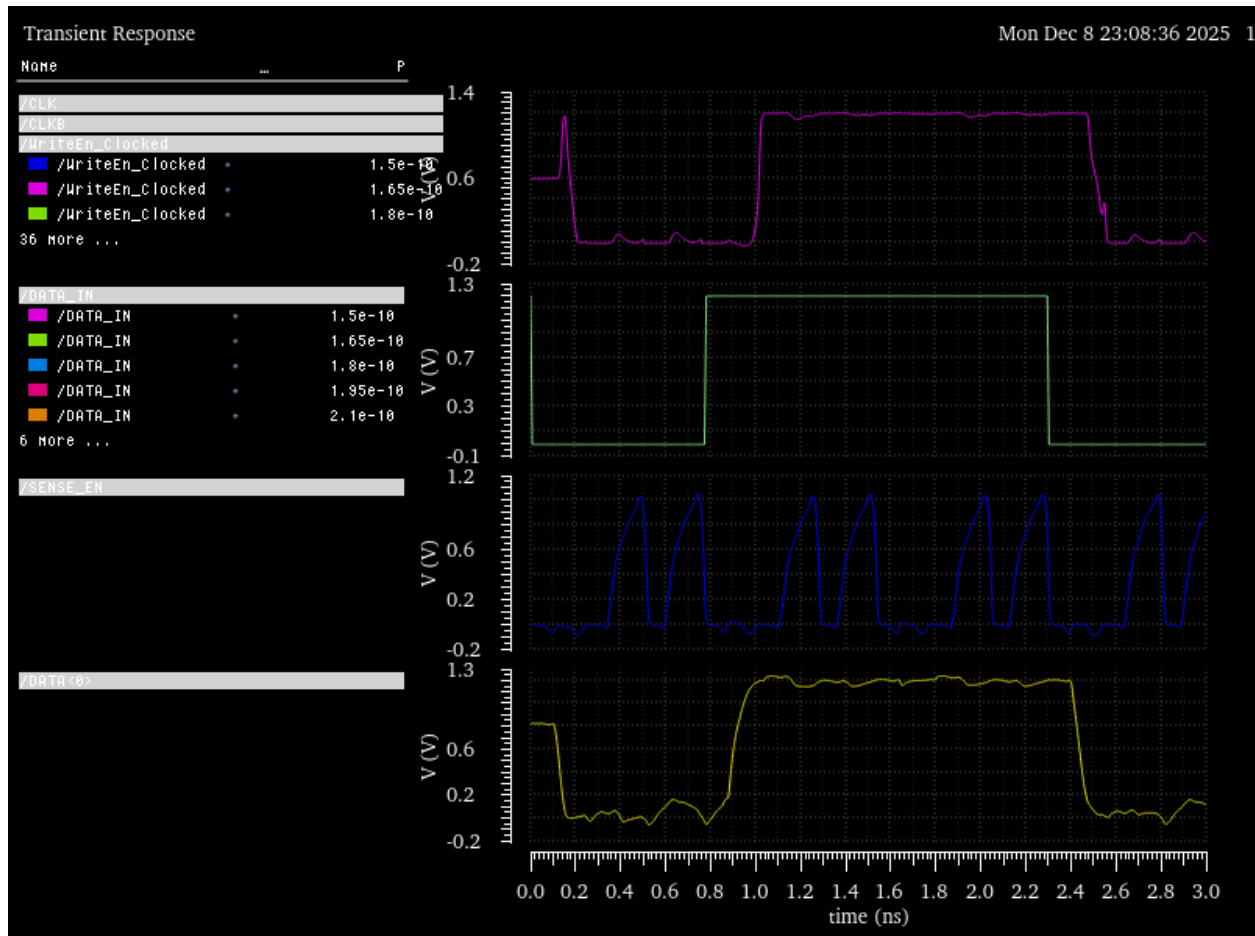


Figure 49. Upon comparing the data bus response to the read enable signal, it is clear that the data does return to the correct output at the very end of the clock cycle, signifying that this period yields correct operation.

With the minimum period established, the average power draw can be measured. Figure 41 shows the supply current during successive write operations. Z

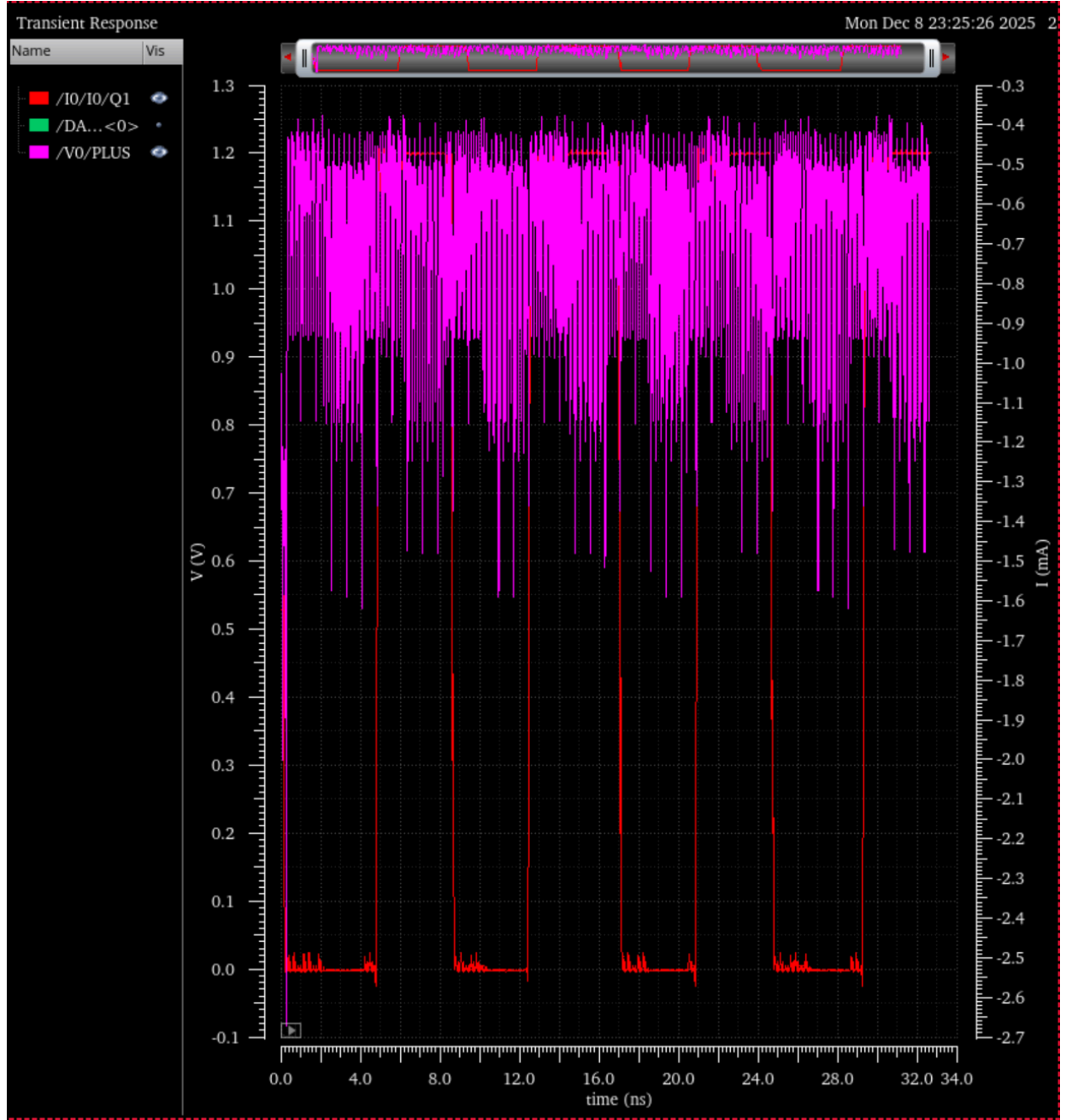


Figure 41. Current draw during the successive write operations performed on the sram cell.

Eqn. 6 is used again to compute power draw, the resultant power draw of the optimized array was determined to be 0.805 mW.

The area of the area is assumed to be the total area of the SRAM cells. Eqn. 7 determines the area of the optimized array.

$$Area = 16 \times 4 \times [(120nm \times 2) + (120nm \times 2) + (120nm \times 2)] \times 45nm = 2.07\mu m^2$$

Using Eqn. 1, the FOM for the optimized array can be determined:

$$FOM = 2.07\mu m^2 \times 0.805mW \times 255ps = 425\mu m^2mWps$$

4.5 FOM Comparison

Each calculated metric, along with the final FOM, is displayed in Table 3.

Table 3. FOM figures for the baseline and optimized designs.

Design	Area (μm^2)	Energy (mW)	Delay (ps)	FOM (μm^2mWps)
Baseline	7.6	0.713	250	1355
Optimized	2.07	0.805	255	425

As seen in Table 3, the optimized design had an FOM about three times lower than that of the baseline design.

5. Conclusion

The design and verification of both the baseline and optimized 64-bit SRAM arrays demonstrate the impact of peripheral circuitry, device sizing, and bitline architecture on overall memory performance. The baseline design, built around a conventional full-VDD precharge scheme and tri-state readout, provides robust functionality and illustrates the classic tradeoffs among read stability, writeability, power, and area. The optimized design introduces a VDD/2 precharge network, minimum-sized 6T cells, and OTA-based sense amplifiers, significantly improving read sensitivity and reducing area while maintaining reliable operation. Although the optimized array exhibits slightly higher dynamic power due to the sense amplifier biasing, the reductions in area and access time dominate the Figure of Merit. Comprehensive simulations confirm that both architectures satisfy the required functionality across all read and write cycles, and the optimized design achieves notably better efficiency according to the adopted FOM formulation. These results highlight the importance of co-optimizing cell devices and periphery when designing high-performance SRAM arrays.